

CHAPTER 1

INTRODUCTION

This chapter serves as a beginning for this research by documenting the problem statement as a inspiration for the development of the project. To be more precise, the sections such as research background, problem statement, research objectives, research scope and research significance will elaborate on such matters.

1.1 Research Background

Recent Artificial Intelligence (AI) advancements have changed today's analysis towards medical images like chest X-ray. Convolutional Neural Networks (CNNs) is one of the Deep Learning (DL) models that have shown its performance that are worth comparing to radiologist specialist in detecting thoracic pathologies from chest X-rays (Ait Nasser & Akhloufi, 2023; L. He et al., 2025). Conditions such as pneumonia, cardiomegaly, pleural effusion, pneumothorax, atelectasis, and lung masses are able to be automatically identified by these models with high accuracy.

Visual Language Models (VLMs) such as LLaVA-Med and GPT-4V have surfaced as the powerful tools for producing meaningful medical explanations and structured radiology report from the inputs of multimodals (M. Li et al., 2023; Wu et al., 2023). These are beneficial as computer vision and natural language can be connects the gaps between them. And allow AI systems to not only detecting abnormalities but also effectively using medical terminology like a professional.

Most existing systems operate component by itself and lacks interaction with other components despite the prominent AI technological advances. The model like CNNs only detects abnormalities in chest X-rays (Rajpurkar et al., 2017), a VLMs only generates radiology reports independently (Alfarghaly et al., 2021; Yang et al., 2023) and another model only retrieves similart historical cases from databases (Jush et al., 2024). They rarely seen work together in a system even if each one of them have shown their technical abilities with its specific task. Thus, the pratical utility of AI systems were break apart as real-world applications require smooth integration of multiple capabalities of AI model to provide supports for diagnosis.

Furthermore, the untrust towards many AI systems been the challenges because of their nature fo lack of transparency or 'black box'. Gradient-weighted Class Activation Mapping (Grad-CAM) is an Explainable AI (XAI) techniques that can offer visual justifications for the

decisions by highlighting the targeted area in medical images (Selvaraju et al., 2017). To strengthen it, techniques like Facebook AI Similarity Search (FAISS), a vector similarity search engine can enable extensive retrieval from large image collection (Jush et al., 2024).

However, a gap in system research remained as for a combination of CNNs for detection, VLMs for reasoning, XAI for visualization and vector-based retrieval methods into a single modular architecture. To accompany it, a chest X-ray datasets such as CheXpert (Irvin et al., 2019) can provide a high quality and large scale benchmark for that such experiments. This research will exploit this data source to perform and show a modular AI systems that combines detection, explanation, retrieval and documentation by presenting modern AI and Big Data engineering practices in a medical context.

1.2 Problem Statement

Current AI systems for chest X-ray analysis have achieved an impressive performance in individual tasks but they were remained fragmented in their implementation and deployment. Detection models such as CheXNet was the first to achieve radiologist level pneumonia detection but produce classification labels without explanatory output (Ait Nasser & Akhloufi, 2023; Rajpurkar et al., 2017). Report generation systems developed by Alfarghaly et al. (2021) and Yang et al. (2023) can produce fluent radiology reports but operate independently of detection confidence or visual evidence. Case retrieval frameworks enable finding of similar historical cases but rarely connect to generative reasoning (Jush et al., 2024).

This fragmentation creates three key technical challenges. First, lack of composability makes it difficult to build systems where modules for detection, explanation, and retrieval chain together reliably. Components that are developed independently are often use incompatible data format or different coordinate systems for localization. Second, limited scalability occurs because many prototypes demonstrate results on small test sets of a few hundred images but do not show how their methods perform on large datasets, leaving data handling untested. Third, weak explainability continues as black box outputs hinder trust and system debugging, even when accuracy is high (Qiu et al., 2023).

These challenges result in practical limitations for AI deployment in medical imaging. User cannot verify model reasoning when detection systems provide no visual explanation. Text generation systems risk on producing hallucinated findings when disconnected from visual evidence. Retrieved similar cases were also provide no educational values when not linked to diagnostic reasoning. Most critically, there is no widely available academic prototype that demonstrates how to engineer a multimodal AI system integrating CNN detection, visual

explanation through Grad-CAM, vector-based retrieval via FAISS, and natural language reasoning through VLMs within a singular and evaluable architecture.

This research addresses the integration gap by designing and implementing a unified system using chest X-rays as a real-world testbed for AI and Big Data integration. The project demonstrates how modern AI components can be composed into a working prototype that provides detection, explanation, retrieval, and reasoning in a single workflow. By documenting the architectural decisions and technical solutions, this work contributes practical knowledge to the field of AI system engineering with applications beyond healthcare.

1.3 Research Objective

To address the identified problem above, this research aims to develop a modular pipeline AI tools with accurate performance and seamless integration. There are three objectives that this research wants to address:

- a) To analyze an approach and suitable method to integrate multiple Artificial Intelligence models in a modular pipeline that is able to assist medical students.
- b) To design and implement a modular pipeline integrating detection, localization, retrieval, and explanation modules during project timeline.
- c) To evaluate the system's performances and effectiveness in improving diagnostic accuracy via technical and usability testing.

1.4 Research Scope

The scope of this research project is carefully defined to ensure focused development while addressing the core integration challenges. The project scope is divided into in-scope elements that will be included and out-of-scope elements that will be excluded.

The project includes the focus only on chest X-ray with posteroanterior (PA) view. This will be accompanied by six common thoracic conditions which are pneumonia, cardiomegaly, pleural effusion, pneumothorax, atelectasis and lung mass/node for detecting.

The project will also implement four AI models, ResNet-50 for detection, Grad-CAM for visualization, FAISS for retrieval and Groq AI which is a VLMs for explanation.

The system will operate by using a CheXpert dataset containing 5,000 to 7,000 images. To evaluate the system, technical metrics including accuracy, precision, recall and F1-

score for detection accuracy and adding System Usability Scale (SUS) for scoring based on user feedback towards the system.

This project will not includes clinical deployment, regulatory approval, and integration with hospital information systems. The system is a research prototype designed not to serve as a clinical diagnostic tool. User behavior analysis that is beyond basic usability feedback is excluded. Additional medical images such as CT scans, MRI or pediatric cases are also excluded.

1.5 Research Significance

This research may bring implications that are significant for multiple aspects of AI development and application. For AI system engineering aspects, this project could be a successful blueprint for a modular AI system by providing a guidance that is practical for other developers that want to build similar real-world AI applications.

Furthermore, for Big Data practice aspects, this project may show how the vector database like FAISS retrieve collections of real medical image. Efficient indexing structure can provides the idea on how to manage resources when working with a large scale datasets.

Moreover, this research also may validates the strategies of practical XAI using Grad-CAM and VLM for medical context. This combination of visual explanation and medical explanation can show how this multiple XAI approaches can complement each other with transparency. This research could proves on how a research with limited resource can built a integrated system prototype using entirely open resources datasets and models.

1.6 Summary

Summing up all of above, this chapter have construct a foundation for an integrated AI systems development by adresssing current fragmentation in medical image analysis tools. The background have highlighted the advancement and how CNNs, VLMs, XAI and data retrieval remains independent in most implementations. Specific challenges that have identified by the problem statement surfaced from this fragmentation.

Objectives of the research will guide the project on analyzing an approach and suitable method for implementing a modular system and evaluating its system performance through technical metrics and usability testing. Moreover, The scope setted up for this research defines a clear boundaries to maintain focus on the integration challenges.

Rather than treating chest X-ray analysis purely as a medical problem, this research uses it as a data-rich domain for demonstrating modern AI system design principles. By implementing a pipeline that unifies detection, visual explanation, case retrieval, and natural language reasoning, the project contributes to the growing body of work on composable and evaluable AI systems with implications extending beyond healthcare.

CHAPTER 2

LITERATURE REVIEW

This chapter will examine and analyze the concept of chest x-ray and each of the Artificial Intelligence (AI) modules by synthesizing it into a literature review.

2.1 Introduction

Chest radiography was originally discovered accidentally by Wilhelm Röntgen in November 1896 while he experimenting with cathode ray tubes in his laboratory in Würzburg, Germany (Mould, 2018). A fluorescent screen across his room began to glow when his cathode ray tube was operating while Röntgen observing it. Soon he realized it was a new type of radiation that could penetrate solid objects. A couple of weeks later he produced the first X-ray image of his wife's hand that is clearly showing her bones and wedding ring.

In 1896 at Boston City Hospital, the first chest radiograph was performed by Dr. Francis Williams marking it as the beginning of thoracic imaging (Thomas et al., 2005). It is required for the patients to remain completely still for several minutes being exposed with weak radiation produced by gas-filled tubes. The images were captured on glass photographic plates coated with light sensitive emulsions as it made them fragile and difficult to store (Mould, 2018).

Time goes by as film-based radiography replaced glass plates to offer better portability and storage. The development of intensifying screens in the 1920s reduced the requirement of radiation doses by amplifying the X-ray effect on film (Thomas et al., 2005). By 1950s, chest X-ray had become a standard diagnostic tool in hospital over the world.

2.1.1 Overview of Chest X-Ray (CXR)

Chest X-ray (CXR) is one of the most frequently performed diagnostic images for procedure in modern healthcare. This diagnosis provides a two-dimensional projection of the thorax by visualizes its critical structures such as the lungs, heart, mediastinum and bony thorax (Ait Nasser & Akhloufi, 2023). Due to its low cost, wide availability and rapid acquisition, the CXR serves as the important imaging modality for assessing respiratory infections, heart failure, trauma and thoracic malignancies across global healthcare systems (L. He et al., 2025).

The standard diagnostic CXR uses a posteroanterior (PA) projection where the X-ray beam travels from the back of a patient to the front with the heart positioned closer to the

receptor of image. This type of positioning minimizes magnification of the cardiac silhouette, which is crucial for cardiomegaly diagnoses accuracy (Ait Nasser & Akhloufi, 2023). In contrast, anteroposterior (AP) views position that the heart is farther from the detector and that can cause artificial enlargement that possibly lead to false-positive diagnoses. Figure 2.1 and 2.2 illustrates the difference between PA (a) and AP (b) positioning.

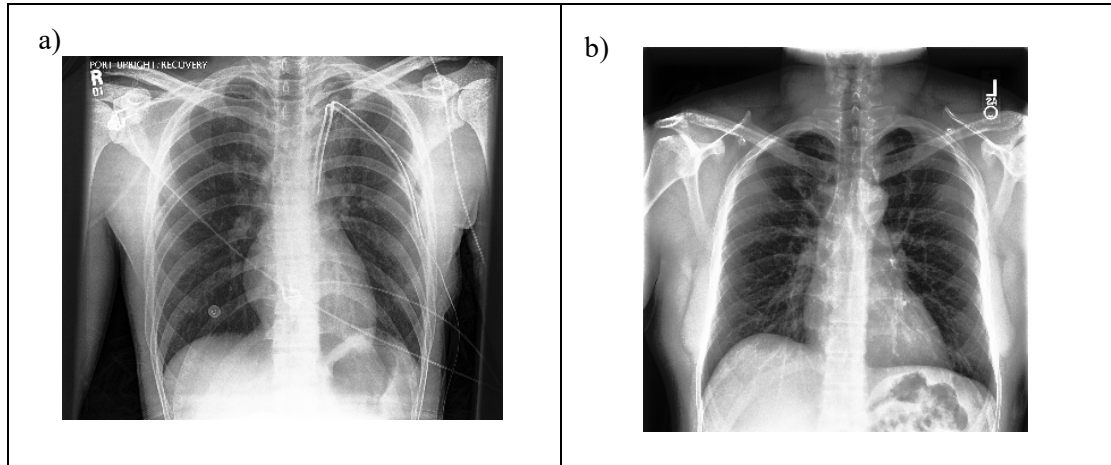


Figure 2.1: AP & PA positioning comparison, retrieved from dataset (Irvin et al., 2019).

The widespread use of CXR across medical departments, intensive care units and outpatient clinics makes accurate interpretation essential for patient care. However, interpretation requires extensive visual training and clinical experience to distinguish subtle abnormalities from normal anatomical variants (Hegazi et al., 2025).

2.1.2 Interpretation Challenges

Medical students and junior doctors commonly struggle with CXR interpretation due to several factors that relate to one another. First of all, there is case exposure limitations during medical rotation and its outcome as access to diverse pathologies varies (Sait & Tombs, 2021). Some students may encounter common conditions like pneumonia repeatedly but they have very few opportunities to see on more rarer findings such as pneumothorax or lung masses.

As for the next, cognitive overload like mental exhaustion affects newcomer who must process multiple aspects of an image at the same time while applying systematic frameworks. Hegazi et al. (2025) had noted that beginner interpreters too attached on obvious findings while overlooked critical subtle signs is a phenomenon known as satisfaction of search error.

Lastly, interpretation errors may go uncorrected by experts for a long time as it due to delayed feedback in traditional learning environments. This delay allows misunderstanding to become more toughen rather than being corrected (Al Elq et al., 2023). Empirical studies have confirm these challenges, with Jimah et al. (2020) founded that confidence levels among

medical students often do not correlate with actual competence in chest radiography interpretation.

2.1.3 Radiologist Shortage Crisis

To make matter worse, the critical global shortage of radiologists have been one of the challenges for education and its particularly in developing countries. In Malaysia, the radiologist per population ratio remains below the World Health Organization (WHO) recommended standard of 4 to 5 radiologists per 100,000 people (Ramli & Mohd Zain, 2023). This shortage problem creates a chain reaction on both patient care and medical education, with fewer experts are available to provide supervised practice and feedback to learners.

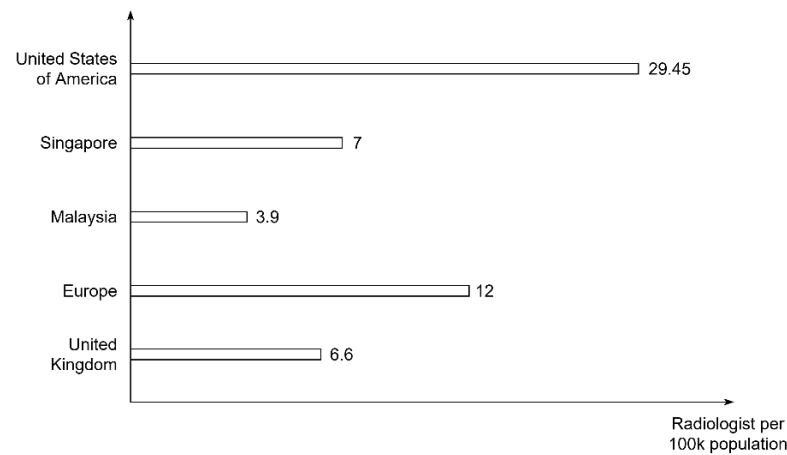


Figure 2.2: Radiologist shortage bar chart across countries, adapted from (Ramli & Mohd Zain, 2023)

The shortage reduces opportunities for bedside teaching and delays expert feedback, slowing the development of pattern recognition skills among trainees. These challenges create a strong rationale for AI-assisted tools that can provide immediate, structured feedback and case-based comparison (Ait Nasser & Akhloufi, 2023).

2.2 Target Pathologies

This research will focuses on six common thoracic pathologies that are frequently encountered in adult chest radiography which is pneumonia, cardiomegaly, pleural effusion, pneumothorax, atelectasis and lung mass or nodule. These conditions were selected based on their clinical commonness, educational connected, and representation in public datasets like CheXpert (Irvin et al., 2019).

2.2.1 Definition and Features

Most of the pathologies definition are based on the infection effect on certain part of the chest. Each of the pathologies will be further explained as what meaning of it and causes it.

2.2.1.1 *Pneumonia*

Pneumonia is an acute infection of the pulmonary parenchyma that leads to inflammation and consolidation (air replaced with fluid, solid or other material). On chest radiography, it show as focal consolidation with air bronchograms that typically in specific lobes (Ait Nasser & Akhloufi, 2023). The condition display high acceptance among radiologists, making it an ideal target for AI-assisted learning.



Figure 2.3: Penumonia on chest X-ray, retrieved from dataset (Irvin et al., 2019).

2.2.1.2 *Cardiomegaly*

Cardiomegaly can be refers to an enlarged cardiac silhouette that is often indicates underlying heart disease. The primary radiographic criterion is a cardiothoracic ratio greater than 0.5 on PA view (Ait Nasser & Akhloufi, 2023). Accurate detection requires distinguishing true cardiac enlargement from artificial magnification in AP views.

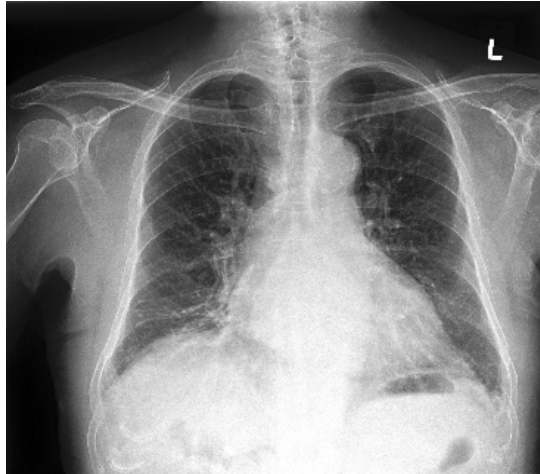


Figure 2.4: Cardiomegaly on chest X-ray, retrieved from dataset (Irvin et al., 2019).

2.2.1.3 Pleural Effusion

Pleural effusion is the accumulation of fluid in the pleural space. Characteristic signs includes blunting of costophrenic angles and the meniscus sign in upright views (Ait Nasser & Akhloufi, 2023). This condition often co-occurs with pneumonia, testing multi-label detection capabilities.

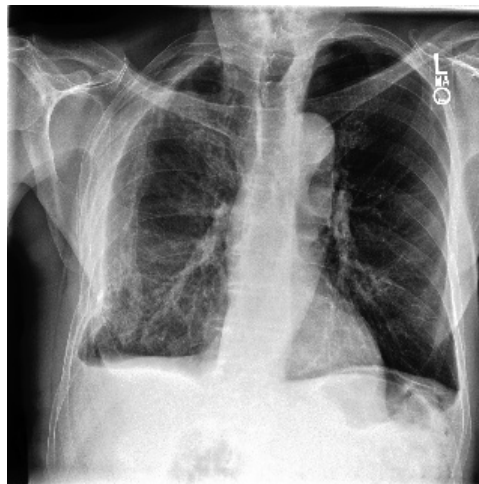


Figure 2.5: Pleural Effusion on chest X-ray, retrieved from dataset (Irvin et al., 2019).

2.2.1.4 Pneumothorax

Pneumothorax means the presence of air in the pleural space that can causes lung to collapse. Diagnostic signs include visible lung edges with absence of peripheral markings (Cho et al., 2021). This represents a critical "cannot miss" finding in emergency radiology.



Figure 2.6: Pneumothorax on chest X-ray, retrieved from dataset (Irvin et al., 2019).

2.2.1.5 Atelectasis

Atelectasis is lung collapse resulting in reduced gas exchange. Unlike pneumonia, it demonstrates the volume loss with elevated hemidiaphragm and crowded bronchovascular markings (Ait Nasser & Akhloufi, 2023). Distinguishing atelectasis from consolidation can be challenging for diagnostic skills.

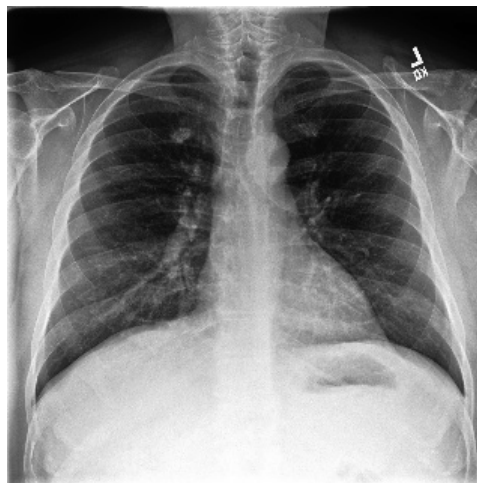


Figure 2.7: Atelectasis on chest X-ray, retrieved from dataset (Irvin et al., 2019).

2.2.1.6 Lung Mass/Node

Lung mass or nodule are abnormal growths in lung tissue, differentiated by size (nodule <3cm, mass >3cm). Malignant lesions typically show spiculated margins, while benign lesions have smooth borders (Ait Nasser & Akhloufi, 2023). These test fine-detail detection capabilities of AI systems.

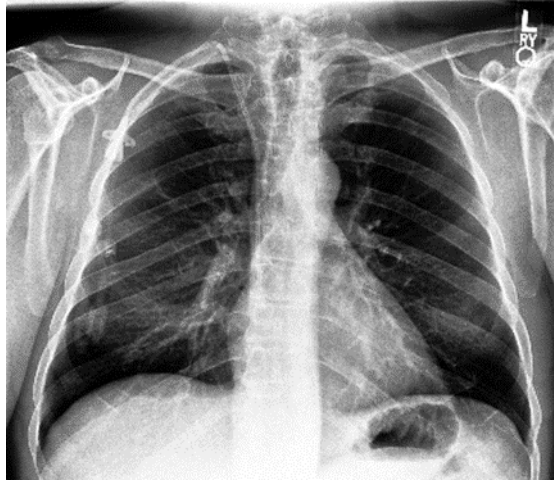


Figure 2.8: Lung Mass/Node on chest X-ray, retrieved from dataset (Irvin et al., 2019).

2.2.2 Comparison of Pathologies

Table 2.1 summarizes the radiographic features and AI relevance of all six pathologies.

Table 2.1

Summary of six pathologies with definitions, key signs, and AI relevance

Pathology	Definition	Key Radiographic Sign	AI Relevance
Pneumonia	Aveolar infection & consolidation	Focal opacity with air bronchograms	High agreement and CheXNet benchmark
Cardiomegaly	Enlarged heart silhouette	Cardiothoracic ratio > 0.5	Tests view orientation awareness
Pleural Effusion	Fluid in pleural space	Blunted costophrenic angle or meniscus sign	Often co-occurs with pneumonia
Pneumothorax	Air in pleural space	Visible lung edge or absent peripheral markings	Critical “cannot miss” finding
Atelectasis	Lung collapse	Opacity with volume loss	Distinguishing from consolidation
Lung Mass/Node	Focal lung growth	Spiculated margins or smooth	Fine detail detection challenge

2.3 Deep Learning (DL)

Convolutional Neural Networks (CNN) have become the foundation for automated chest X-ray analysis replacing traditional feature extraction method. Among numerous architectures emerged, they have being leading candidates for thoracic pathology detection. This section compares these models and justifies the selected model for this research.

2.3.1 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) are specialized deep learning architectures that are designed to process grid-structured data such as images. The fundamental structure of CNNs includes the convolutional layers where learned filters are to scan the input in a sliding window manner for detecting features such as edges, textures, and patterns. These filters can produce a feature maps that will encode the presence and spatial organization of features at different scales (L. He et al., 2025). Pooling layers will follow the convolutional layers to reduce more computational complexity and improve the robustness to differences and patterns observed by downsampling the feature maps. Then, the network concludes with fully connected layers that will combine the extracted features and producing a final classification outputs.

Such a hierarchical structure will allows the CNNs to learn potray progressing from simple patterns in early layers to complex and abstract features in deeper layers(Ait Nasser & Akhloufi, 2023). In medical imaging, CNNs have demonstrated a remarkable success in automating the diagnostic tasks. As for chest X-ray analysis, CNNs can learn to identify pathological patterns directly from pixel values and it eliminating the need for manually crafting new features. The availability of large labeled datasets such as CheXpert has enabled training of sophisticated models that approach or exceed human-level performance on specific tasks (L. He et al., 2025).

Several CNN architectures have emerged as leading candidates for thoracic pathology detection, each with distinct design principles and performance characteristics. The following subsections describe four prominent architectures used in medical imaging research.

2.3.1.1 *ResNet-50*

ResNet-50, introduced by K. He et al. (2016), addresses the vanishing gradient problem in very deep networks through residual connections. The architecture consists of 50 layers organized into residual blocks, where skip connections allow gradients to flow directly through the network during training. This design enables stable training of deep networks, achieving higher accuracy than shallower architectures.

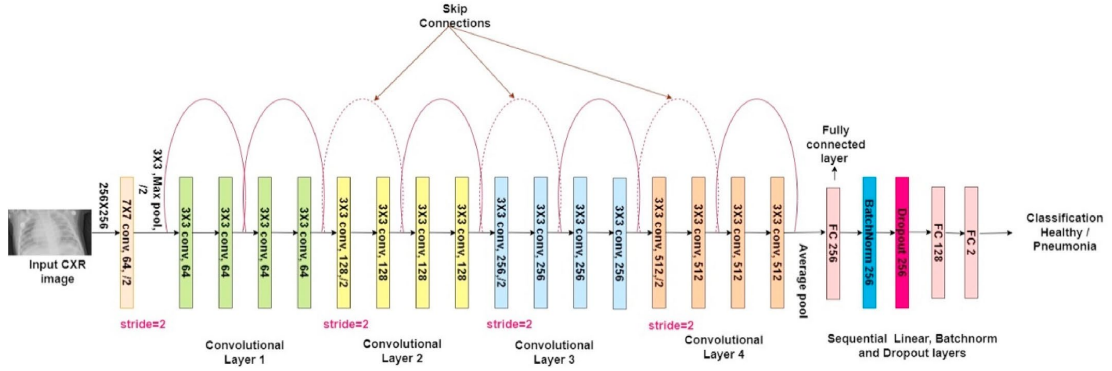


Figure 2.9: ResNet-50 Architecture Diagram, retrieved from (Showkat & Qureshi, 2022).

ResNet-50 affect a balance between model depth and computational efficiency for making it suitable for resource-constrained academic environments (Showkat & Qureshi, 2022). Transfer learning with ImageNet-pretrained weights have shown a high sensitivity in pneumonia detection, with the model working seamlessly with explainable AI techniques like Grad-CAM (L. He et al., 2025).

2.3.1.2 DenseNet-121

DenseNet-121 uses dense connectivity where each layer connects to all subsequent layers within a dense block for maximizing feature reuse (Huang et al., 2017). This architecture allows the network to learn more diverse representations with fewer parameters than traditional CNNs. DenseNet-121 gained its important through CheXNet which a landmark study that achieved radiologist-level pneumonia detection on the ChestX-ray14 dataset (Rajpurkar et al., 2017).

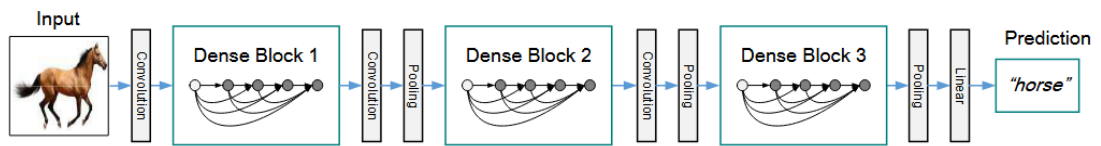


Figure 2.10: DenseNet Architecture Diagram, retrieved from (Huang et al., 2017).

However, dense connectivity comes with computational costs, requiring 20 to 30 percent longer inference time compared to ResNet-50 due to increased memory consumption from concatenating feature maps (Ait Nasser & Akhloufi, 2023). The larger memory footprint makes it less suitable for real-time applications.

2.3.1.3 VGG-16

VGG-16 is developed by Simonyan & Zisserman (2015) has follows a simple and uniform architecture that consists of 16 weight layers with small 3×3 convolutional filters that are stacked one after the another. VGG-16 network depth and use of small filters allowing it to learn more complex feature representations while maintaining computational from being easily persuaded. VGG-16 has demonstrated that by increasing network depth with very small convolution filters leads to significant improvements in image classification accuracy.

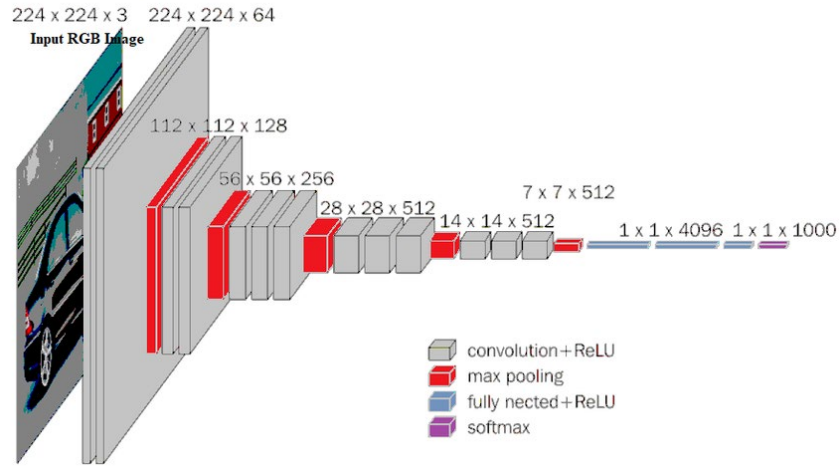


Figure 2.11: VGG-16 Configuration, retrieved from (Simonyan & Zisserman, 2015)

In medical imaging, VGG-16 has been applied to various diagnostic tasks and showing a strong performance in feature extraction (Bharati et al., 2020). However, its architecture requires substantial memory and computational resources due to large number of parameters that is approximately 138 million making it less practical for deployment in limited resource of hardwares compared to more modern architectures.

2.3.1.4 EfficientNet-B0

EfficientNet-B0 uses neural architecture search to optimize the compromises between accuracy and computational efficiency (Tan & Le, 2020). This model is systematically scales network depth, width and resolution by using a compound scaling method that can balances all three dimensions. This approach would achieves much more higher accuracy with fewer parameters and lower computational cost as compared to traditional architectures.

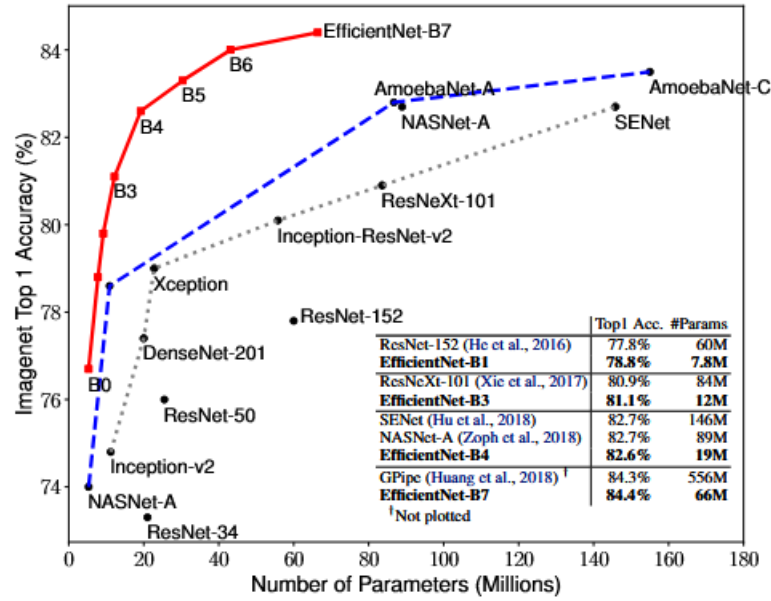


Figure 2.12: EfficientNet-B0 accuracy by number of parameters, retrieved from (Tan & Le, 2020).

Moreover, EfficientNet-B0 is particularly suitable for edge devices and mobile applications where computational resources are limited. However, its adoption in medical imaging remains limited as compared to ResNet and DenseNet with fewer validated studies that specifically demonstrating its effectiveness for chest X-ray analysis.

2.3.2 Comparison of Deep Learning Model

Table 2.2 provides a complete comparison of the four CNN architectures across their key performance and practical considerations that are relevant to this research.

Table 2.2
Comparison of CNN architectures

Model	Depth	Parameters	Strength	Limitation	Medical Validation	References
ResNet-50	50 layers	25.6M	Fast inference, Grad-CAM compatible, strong transfer learning	Slightly lower accuracy than DenseNet	Extensive	(K. He et al., 2016; Showkat & Qureshi, 2022)
DenseNet-121	121 layers	8.0M	High feature diversity, radiologist-level accuracy	Slow inference (20-30% slower), memory-intensive	Extensive	(Rajpurkar et al., 2017)

VGG-16	16 layers	138M	Simple architecture, strong feature extraction	Very high parameter count, large memory footprint	Moderate	(Bharati et al., 2020)
EfficientNet- B0	Variable	5.3M	Optimized efficiency, smallest model size	Limited medical imaging validation	Limited	(Tan & Le, 2020)

As for the summary of the comparison, ResNet-50 appears as the most optimal choices for this research based on several factors. Firstly, ResNet-50 computational efficiency allows fast prototyping and real time inference on a standard software. Secondly, ResNet-50 maintains the compatibility with Grad-CAM for explainability and FAISS for vector-based retrieval which are both a critical components of the integrated system. Lastly, the existence of large community support ensures the ResNet-50 reproducibility.

While the compared model like DenseNet-121 may achieve slightly higher Accuracy scores in benchmarks studies, ResNet-50 provides a sufficient accuracy for implementing an AI system integration. This research is emphasizes on composable system engineering rather than maximizing single-task performance. By making ResNet-50's balance of speed, compatibility and academic feasibility most appropriate (L. He et al., 2025).

2.4 Explainable AI (XAI)

Just as convincing was deep learning with its high accuracy in chest X-ray analysis, the lack of transparency from their nature have limits the trust and utilization for medical purposes. To solve this, Explainable AI (XAI) techniques address this by providing visual justifications for the CNNs by pointing where the model focuses when making a diagnosis. Such capability is crucial in medical education where students have to understand what is the reason behind the evidence of why and where.

2.4.1 The Black Box Problem

The problem originates from the operation of traditional deep learning models which they providing prediction without explaining the reasons of that processes. Eventhough the model might correctly identified the pathologies in a X-ray, its challenges is providing without an evidence the regions influencing the decisions. This effects users who cannot verify whether

the model use relevant signs or relies on fake correlations. This lack of transparency creates several problems in medical context (Qiu et al., 2023).

At first, doctors cannot trust a prediction that cant be verified yet. A radiologist wants a confirmation about the AI diagnosis that the model identified and not getting distracted by random artifacts in the photo such as label and metal marker. Second, learners cannot develop diagnostic skills by observing unexplained predictions. Medical education requires on understanding the visual patterns that distinguish each pathology, which black box models fail to provide (Plackett et al., 2022). Third, regulation and liability concerns arise when AI systems make high-stakes medical decisions without interpretable reasoning. Healthcare providers need to justify their diagnostic conclusions which can becomes impossible when relying on not transparent AI systems (Plesner et al., 2023).

These limitations motivate the development of XAI techniques that can visualize model attention and provide interpretable explanations. For educational AI systems, explainability transforms the tool from a mere decision-maker into a teaching instrument that demonstrates diagnostic reasoning through visual evidence. Several XAI methods have been developed to address the black box problem that each of it with different approaches to visualization and interpretation.

2.4.1.1 Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) generates class-discriminative heatmaps by computing the gradients of a target class score with respect to the final convolutional layer feature maps (Selvaraju et al., 2017). This method identifies which of the regions of an image most strongly influence the model's prediction for a specific class.

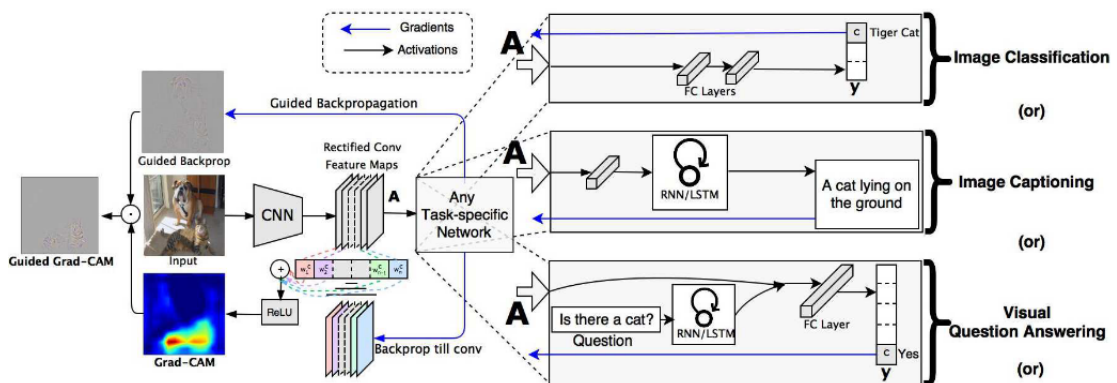


Figure 2.13: Grad-CAM architecture overview, retrieved from (Selvaraju et al., 2017).

In medical imaging, Grad-CAM's strength lies in its clinical interpretability, consistently highlighting anatomically relevant regions such as the lung edge for pneumothorax

or affected lobes for pneumonia (Qiu et al., 2023). Importantly, Grad-CAM requires no architectural changes to the base model, making it ideal for transfer learning scenarios. The coarse but anatomically meaningful heatmaps provide sufficient guidance to direct attention to relevant anatomical areas while encouraging active examination of images.

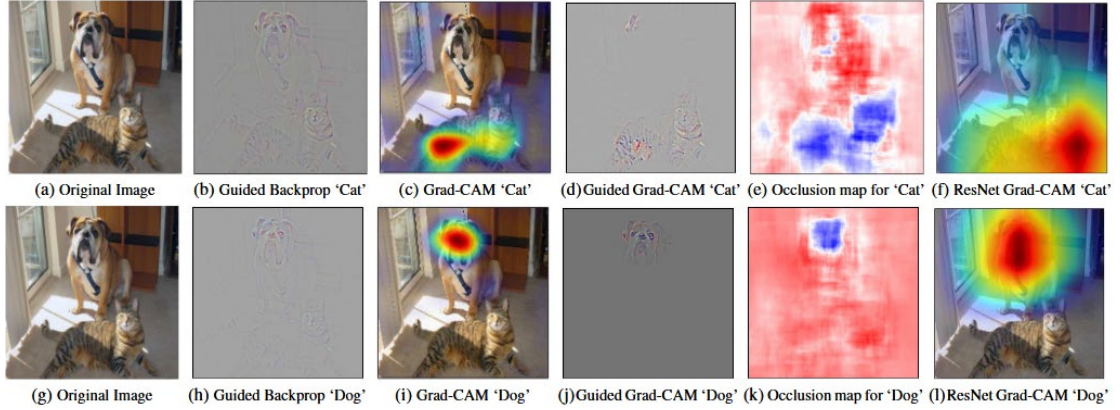


Figure 2.14: Grad-CAM in examining distinct features, retrieved from (Selvaraju et al., 2017).

2.4.1.2 Saliency Maps

Saliency Maps compute the gradient of the output class with respect to input pixels, producing pixel-level sensitivity maps that show how small changes to each pixel would affect the prediction (Simonyan & Zisserman, 2015). While this provides a fine-grained detail, saliency maps often yield noisy and small grains outputs that lack clinical coherence. Qiu et al. (2023) have compared Grad-CAM and Saliency Maps on chest X-rays and found that saliency maps frequently highlight random texture artifacts rather than meaningful pathology. This unreliability makes them unsuitable for educational use where conceptual clarity is more important than anything else. Students presented with noisy visualizations may become confused about which features are relevant to the diagnoses.

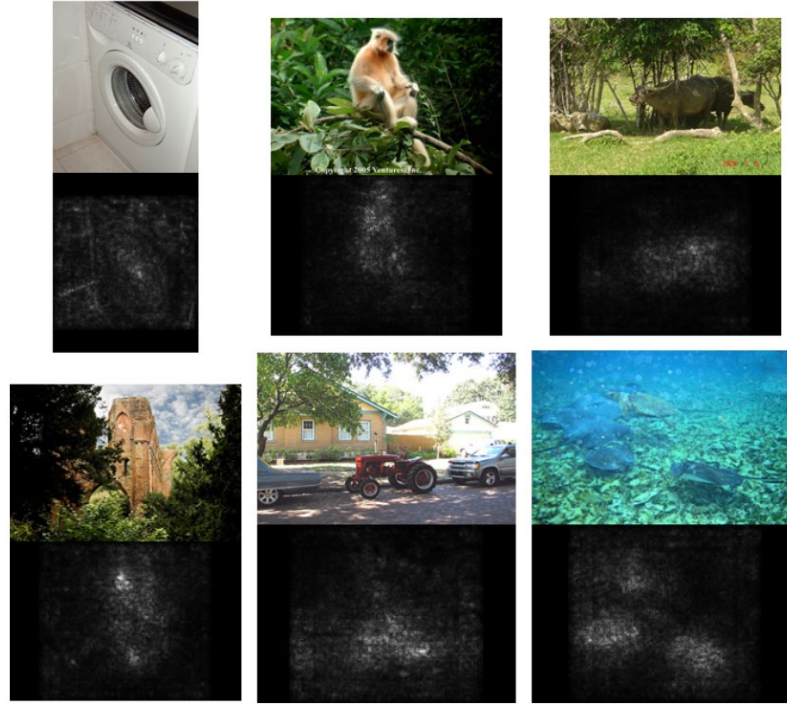


Figure 2.15: Saliency Maps output, retrieved from (Simonyan & Zisserman, 2015).

2.4.1.3 Attention Mechanism

Attention mechanisms are popularized by Vision Transformers able to integrate interpretability directly into model architecture. Each attention centre the focuses on specific image regions, and these attention weights can be visualized as heatmaps (Shin et al., 2020). However, attention-based XAI requires custom architectures and extensive retraining, which is unfeasible for academic prototypes using pretrained CNNs. Moreover, attention maps reflect model training dynamics rather than necessarily indicating clinical relevance that potentially misleads learners about diagnostic reasoning. The computational cost and limited validation in medical imaging contexts restrict their practical application in educational settings.

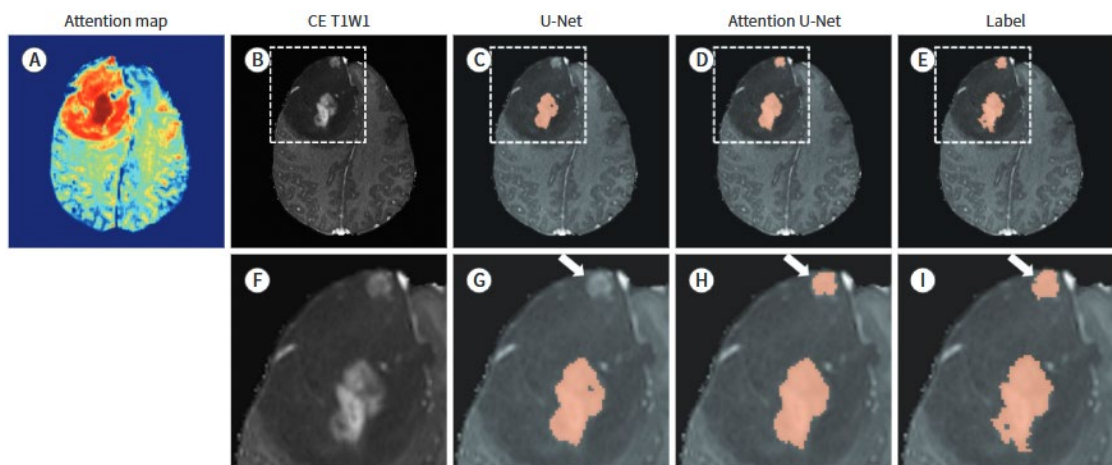


Figure 2.16: Attention Mechanisms overlaid on brain image, retrieved from (Shin et al., 2020).

2.4.2 Comparison of XAI Techniques

Table 2.3 compares the three XAI techniques across key criteria for educational applications.

Table 2.3
Comparison of XAI techniques

Method	Mechanism	Resolution	Strength	Limitation	Medical Validation	References
Grad-CAM	Gradients towards final convolutional layer	Coarse	Clinically interpretable, model-agnostic, lightweight	Lower spatial precision	Strong	(Qiu et al., 2023; Selvaraju et al., 2017)
Saliency Maps	Gradients towards input pixels	Fine (pixel-level)	High spatial detail	Noisy, unstable, highlights artifacts	Weak	(Qiu et al., 2023)
Attention Mechanisms	Transformer attention weights	Variable	Integrated into architecture	Requires custom models, expensive retraining	Limited	(Shin et al., 2020)

Grad-CAM emerges as the most suitable XAI method for this system based on several criteria. First, its clinical reliability has been validated across multiple studies showing alignment with expert radiologist attention patterns (Qiu et al., 2023). Second, its model-agnostic nature allows it to work with any CNN architecture, including the pretrained ResNet-50 selected for this research. Third, the computational overhead is minimal, adding negligible processing time to inference.

Fourth, the coarse but anatomically meaningful heatmaps serve an educational purpose by guiding learners to correct anatomical regions while requiring active examination. Finally, Grad-CAM's widespread adoption in medical AI research ensures that the system's explainability approach is familiar to the broader research community, facilitating validation and comparison with other studies.

2.5 Vector-Based Case Retrieval

A core educational strategy in radiology training like Case-Based Reasoning (CBR) where students can improve their diagnostic accuracy by comparing with other verified cases. However, metadata tagging or pixel based similarity is a traditional image retrieval methods

that are fail to capture the meaning of visual patterns that define medical pathologies. Deep learning is the modern systems that uses images converting into high dimensional embeddings and then compare the vector using similarity metrics. This enables the system to find past cases that are similar even if they aren't labeled as the same.

2.5.1 Case-Based Reasoning (CBR)

Case-Based Reasoning (CBR) is a learning pattern where students are allow to develop diagnositic skills by comparning and examining multiple examples that matches their conditions. When unclear chest X-ray are encountered by a students, they can access verified similar cases that provides context that helps in patterns recognition and diagnostic confidence. Sugi et al. (2021) have demonstrated interactive and case-based learning can improves radiology education outcomes as compared to traditional leacture approaches.

CBR can be effectives if it relies on the quality and relevance of retrieved cases. While the manually curated of similar consumes time and subjective, it can depends on how radiologists categorize and tag the images. Distinguishing subtle diffrences in visual appearances cannot by matching simple keywords as it limits the educational value of the retrieved examples. The way of modern systems is by using deep learning to convert the images into high-dimensional embeddings and then compare its vectors using similarity metrics. This enables retrieval of similar cases based on the actual appearances rather than manual annotations (Jush et al., 2024). Hence, by using learned image representations that capture visual similarity, semantic retrieval had addresses this challenges beyond categorical labels (Al Elq et al., 2023).

2.5.1.1 Facebook AI Similarity Search (FAISS)

Facebook AI Similarity Search (FAISS) is an optimized open-source library for efficient similarity search in high-dimensional vector spaces (Johnson et al., 2021). It supports both exact search using brute-force distance computation and approximate search using indexing techniques such as Inverted File (IVF) and Product Quantization for improved speed. The library includes GPU acceleration, enabling real-time retrieval from large databases.

In medical imaging, Jush et al. (2024) demonstrated that FAISS with ResNet-50 embeddings achieves high recall in chest X-ray retrieval tasks, successfully retrieving visually and clinically relevant cases from datasets like CheXpert. FAISS achieves sub-second retrieval times on datasets containing thousands of images, requires no separate database server as embeddings are stored as binary files, and handles future dataset growth through scalable indexing methods.

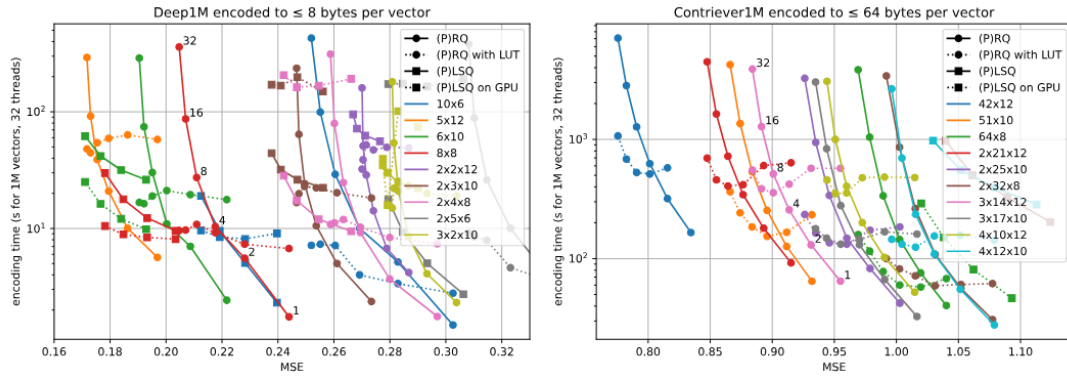


Figure 2.17: FAISS accuracy, retrieved from (Douze et al., 2025).

2.5.1.2 Annoy

Annoy (Approximate Nearest Neighbors Oh Yeah), developed by Bernhardsson (2013), uses tree-based indexing for approximate nearest neighbor search. The method builds multiple random projection trees during the indexing phase, then queries these trees to find candidate nearest neighbors efficiently. While lightweight and simple to implement with a straightforward API, Annoy is slower than FAISS at scale that typically requiring 1 to 2 seconds for queries on datasets of 5,000 images. Additionally, Annoy lacks GPU support that making it less suitable for real-time applications where responsiveness is critical. The method has not been covers largely to validate in medical imaging literature, limiting confidence in its reliability for clinical or educational contexts.

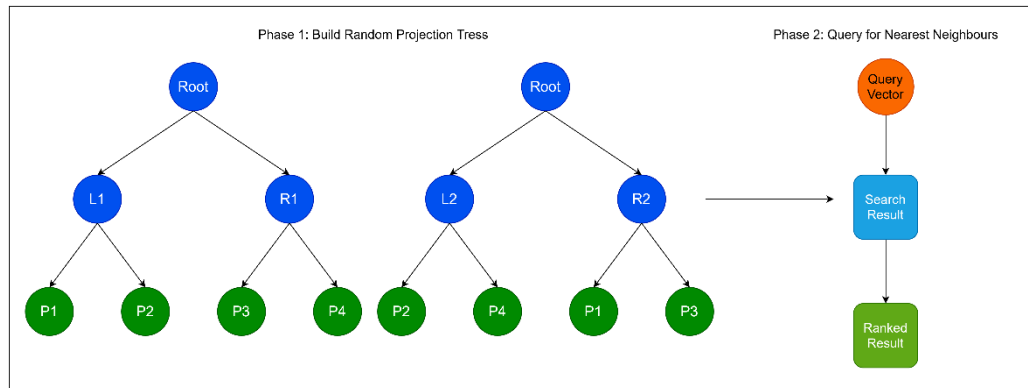


Figure 2.18: Annoy architecture overview, adapted from (Bernhardsson, 2013).

2.5.1.3 Elasticsearch

Elasticsearch by Elastic (2023) are accompanied by vector plugins who supports hybrid text and vector search. It allows multimodal queries that had keyword combination with visual similarity. For an instance, a querie might states specifically with “pneumonia in the right lower lobe” while also finding obvious similar cases. Elasticsearch’s distributed architecture has been leveraged by this approach for exceptional scalability and rapid query times that’s under 0.3

seconds. However, server infrastructure and complex configuration are required by Elasticsearch showing deployment overhead for a prototype. This impractical need for server management, security configurations and system administration for limited resource research where simplicity and ease of deployment are the priorities.

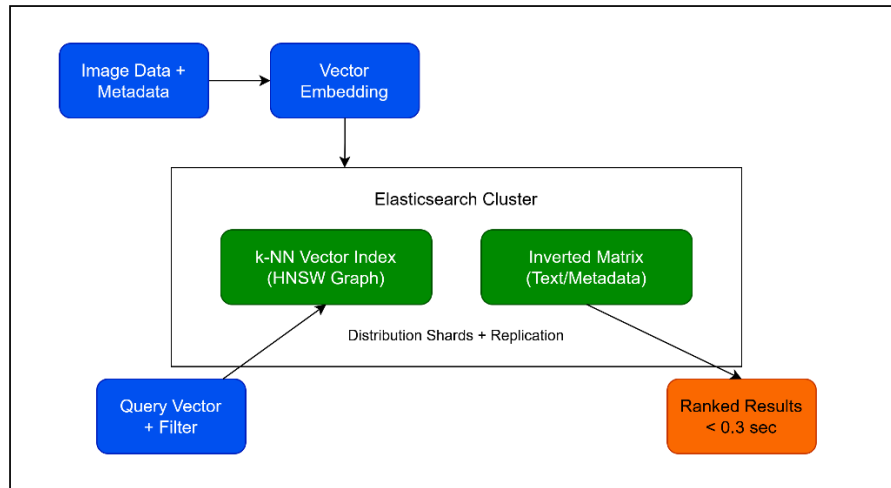


Figure 2.19: Elasticsearch Vector Search Architecture, adapted from (Elastic, 2023).

2.5.1.4 Traditional Metadata Search

Traditional metadata can be considered as categorical labels that relied upon such as “pneumonia = 1” or “atelectasis = 0” which is stored in a database tables. Simple SQL WHERE clauses that used by queries is for filter cases by label combinations. Even if its implementation are fast and simple, there is severe limitation for this approach. The search ignores visual appearance entirely and distinguish between subtypes or severity levels within same category cannot be made. Metadata search treats every cases as equally similar even there’s different in patterns and location. This lack of understanding put the traditional metadata search as poor for case-based learning where visual similarity is more important.

2.5.2 Comparison of Retrieval Method

Table 2.4 compares the four retrieval methods across key criteria relevant to educational applications.

Table 2.4
Comparison of retrieval methods

Method	Semantic Understanding	Speed	Scalability	Implementation Complexity	Medical Validation	References
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FAISS	High (deep features)	<0.5 sec	Excellent (GPU support)	Low (Python library)	Yes	(Johnson et al., 2021; Jush et al., 2024)
Annoy	High	1-2 sec	Good	Low	No	(Bernhardsson, 2013)
Elasticsearch	High (with vector plugin)	<0.3 sec	Excellent	High (server setup)	Limited	(Elastic, 2023)
Traditional Metadata	Low (label-only)	<0.1 sec	Poor	Very low	Implied limitation	(Sugi et al., 2021)

As for the summary of the comparison, FAISS have emerges as the optimal choice for this research based on several factors. First of all, FAISS achieves the best balance of speed and semantic understanding. Second, excellent scalability can be provided by GPU acceleration and efficient indexing for future growth of dataset without requiring changes on the architecture. Thirdly, FAISS is lightweight Python library by eliminating complex server infrastructures as it crucial for prototype with limited resources.

Fourth, scientific evidence in medical imaging literature (Jush et al., 2024) could provides confidence in its reliability for chest X-ray retrieval. While Elasticsearch offers slightly faster queries, its implementation complexity and server requirements make it impractical for this research scope. Despite the simplicity of traditional metadata search, its has fails to capture the necessary visual similarity for effective case-based learning.

2.6 Visual Language Models (VLMs)

As detection provide ‘what’ and localization provide ‘where’, the understanding of ‘why’ are required for medical education. Thus, Visual Language Models (VLMs) steps in to address this by creating natural language reasoning that imitate expert diagnostic. This model can answer questions such as “Why does this detected as cardiomegaly?” or What characteristics is the pneumonia in this?” that aligns with computer vision and natural language processing (Yang et al., 2023).

2.6.1 Multimodal AI

VLMs poses as an evolution in AI systems that process and generate information using computer vision and natural language processing from both images and text at the same time. VLMs would connects the gap in medical imaging between visual language and linguistic explanation. This can achieve by enabling systems to not only detect pathologies by also outspoken their reasoning in medical meaningful language (Yang et al., 2023). This potential is valuable for applications where learners needs to understand presented abnormalities and reasons for specific diagnosis conclusion are reached.

Old report generation systems that relied on template based have been left behind from the appearance of VLMs. The earlier systems had to used conditioned transformers to generate radiology reports but they sometimes produces text that did not accurately described as the image content (Alfarghaly et al., 2021). Meanwhile, VLMs uses vision and language with attention mechanism that roots it explanation from specific visual features. This keep the generated text to corresponds with actual image content rather than solely from learned patterns.

VLMs has the potential to offer reasoning to aid students in understand logic of diagnosis. Rather than simply stating the detection results such as “pneumonia detected”, VLM can explain a full sentence of impression on the images. This structured explanations guide students to be attached with visual observations and diagnostic summary for supporting the medical reasoning skill development.

2.6.1.1 *Groq API (Cloud-based)*

Groq provides cloud-based API access to vision-capable language models optimized for fast inference through specialized hardware acceleration (Groq, 2024). The service offers simple API integration without requiring local GPU resources or model management, making it accessible for academic prototypes with limited computational infrastructure. Groq's architecture emphasizes inference speed, achieving response times of 3 to 5 seconds for chest X-ray report generation tasks.

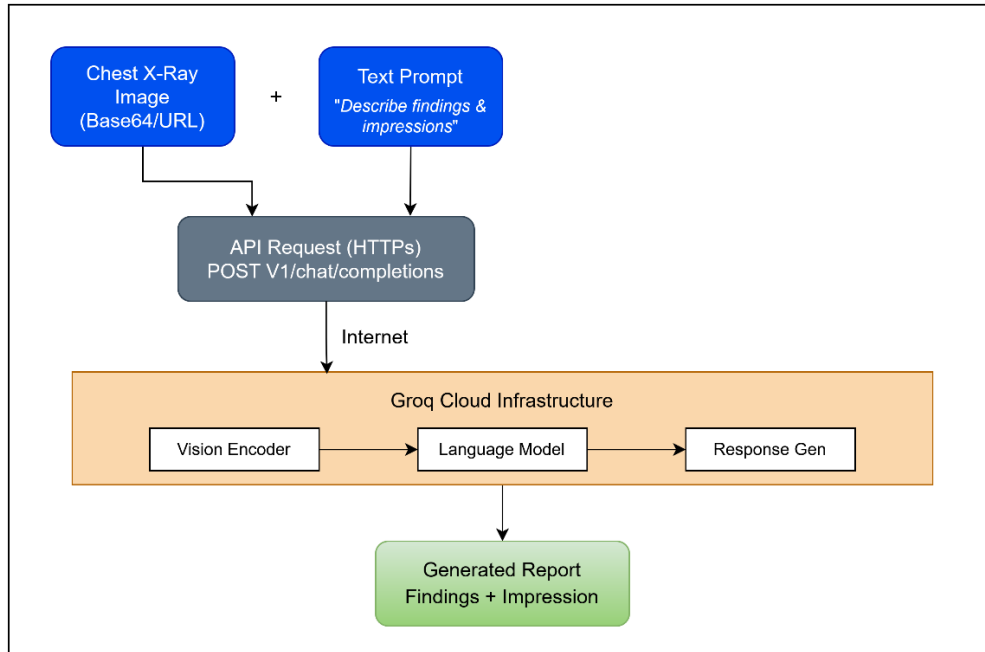


Figure 2.20: Groq API pipeline, adapted from (Groq, 2024).

The API supports multimodal inputs including images and text prompts to enabling zero-shot prompting strategies without fine-tuning requirements. However, as a cloud-based service, Groq requires a stable internet connectivity and involves per-request costs but through free tier access is available for research purposes. The ownership nature of the underlying models limits transparency about training data and model architecture.

2.6.1.2 LLaVA-Med (Local Deployment)

LLaVA-Med (Large Language and Vision Assistant for Medicine) is a biomedical-specialized VLM developed through fine-tuning on over 10 million image-text pairs from biomedical literature (C. Li et al., 2023). Unlike models that are general-purpose, LLaVA-Med has been trained specifically on medical terminology, anatomical concepts and clinical reasoning patterns. The results is with improved accuracy and appropriateness for medical applications.

The key advantages for LLaVA-Med are including open-source availability allowing local deployment without relying on commercial cloud service, understands with medical terminology such as "air bronchograms" and "spiculated margins" and support for customized exclusively prompts that encourages reasoning of the diagnoses step-by-step. Local deployment provides full control over model execution and data privacy, with no per-request costs after initial setup. However, running LLaVA-Med requires substantial computational resources,

including approximately 13GB storage for model weights and GPU memory for inference, making it less practical for resource-constrained environments.

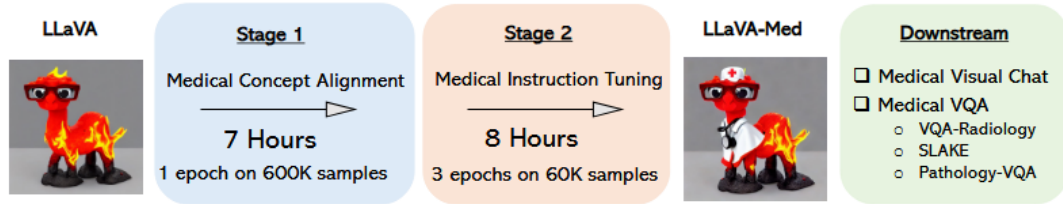


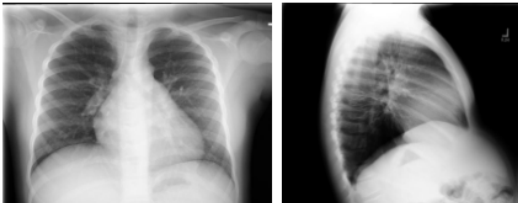
Figure 2.21: Multimodal biomedical understanding examples, retrieved from (C. Li et al., 2023)

2.6.1.3 GPT4-V (Cloud-based)

GPT-4V (GPT-4 with Vision) is a multimodal extension of OpenAI's GPT-4 language model that accepts both image and text inputs. Wu et al. (2023) have evaluated GPT-4V on various medical imaging tasks and found it demonstrates a remarkable zero-shot reasoning capabilities for generating fluent and structured reports without task-specific fine-tuning. The model can interpret chest X-rays, identify abnormalities and provide diagnostic suggestions using natural language.

Cardiac: Case 1

Prompt
Please generate a structural report according to these images.



GPT-4V

Findings:

- Lungs and Pleura:**
Both lungs show clear and well-inflated lung fields without obvious consolidation, mass, or pleural effusion.
No visible pneumothorax.
- Cardiomediastinal Contour:**
The mediastinal and cardiac contours appear within normal limits.
- Bones and Soft Tissue:**
Visualized osseous structures of the chest show no obvious fractures or lytic lesions.
The soft tissues are unremarkable.
- Other Observations:**
The provided images are slightly rotated as evidenced by the misalignment of the vertebral spinous processes with respect to the trachea. This may affect the accurate assessment of the cardiac size and mediastinal structures.

Impression:

- No acute cardiopulmonary abnormality detected on the provided images.
- Consider clinical correlation.

Reference answer

- Mild diffuse cardiomegaly with mild central pulmonary vascular congestion in this patient with sickle cell disease.
- Mild diffuse bilateral bronchial wall thickening may be due to reactive airway disease or a chronic finding. Minimal fibro-atelectasis in the right mid lung zone. No focal pneumonia, effusion or pneumothorax. Lungs otherwise clear.
- Mild anterior wedging and biconcavity of some thoracic vertebral bodies, mild diffuse endplate sclerosis and H shaped configuration of the lower-thoracic vertebral bodies on frontal projection.
- Diffuse splenomegaly in the upper abdomen best evident on the lateral view.

Figure 2.22: Examples of radiology report generation, retrieved from (Wu et al., 2023).

However, GPT-4V is suffer to hallucinations where it invents findings that are not present in the image such as reporting pleural effusion when none exists or describing lesions in anatomically incorrect locations (Wu et al., 2023). These errors pose serious risks in educational settings where factual accuracy is essential. Additionally, GPT-4V operates as a proprietary commercial service with API costs, lacks specific biomedical fine-tuning compared to domain-specific models, and requires internet connectivity for all requests.

2.6.2 Prompt Engineering

Fine-tuning VLMs can be very rewarding but its requires substantial computational resources and labeled data. This research uses zero-shot prompt engineering to guide model outputs. Prompt engineering involves carefully designing the text instructions provided to the model to elicit desired response formats and content. Yang et al. (2023) has demonstrated that structured prompts improve the quality and clinical relevance of generated radiology reports as compared to simple open-ended questions.

The system uses two part of prompting approach that separates observation from interpretation. First part is the Findings prompt, it instructs the model to describe about the findings in the chest X-ray in about 1 to 2 sentences with using medical terminology and not include a diagnosis. The reasons is to encourgaes the model to focuses on observable features without conclusions such as opacities, cardiac size and etc.

The second part is Impression prompt, its ask to provide the diagnosis based on the findings. This structured approach strengten the medical aligned output to matchin professional diagnose report standards. Hallucinations can be reduced by rooting the resoning in explicit findings before allowing diagnostic conclusion. Education that teaches students to separate observation from an intepretation can also be provided. The prompts also can be incorporate detection labels and retrieval results. This would creates a retrieval augmented generation strategy that roots VLM reasoning in verified information (Yang et al., 2023).

2.6.3 Comparison of VLMs approaches

Table 2.5 compares the three VLM approaches across key criteria for educational applications.

Table 2.5
Comparison of VLM approaches

Model	Deployment	Training Data	Strength	Limitation	Cost	Reference
Groq API	Cloud	General (not medical-specific)	Fast inference (3-5 sec), simple integration, no local GPU needed	Proprietary, requires internet, per-request costs	Free tier + paid	(Groq, 2024)
LLaVA-Med	Local	10M+ biomedical image-text pairs	Biomedical knowledge, open-source, data privacy, no per-request costs	Requires 13GB storage + GPU, slower inference (8-12 sec)	Free (one-time setup)	(C. Li et al., 2023)
GPT-4V	Cloud	General (not medical-specific)	Fluent reasoning, zero-shot capability	Prone to hallucinations, expensive API, no biomedical fine-tuning	Paid only	(Wu et al., 2023)

For the conclusion of the comparison, Groq API appears as the optimal choice for this research based on several factors. First of all, its cloud-based deployment eliminates the need for local computational resources as this could ease the prototype development as its reduce the time for setting up the model. Second, Groq API have fast inference times of 3 to 5 seconds responsive user interactions compared to 8 to 12 seconds for local models. Thirdly, simple API integration reduces implementation complexity and allow to focus on system integration rather than model management.

Fourth, Groq API is cost effective as the free tier access for research purposes without going for expensive and outgoing bills or subscriptions. Compared to LLaVA-Med who offered medical specialization, its so impractical if the research have limitation of resource. Compared to GPT-4V, its tends to hallucination and high costs without medical specialization make it less suitable despite its fluent language generation.

2.7 Research Gap

Most of the studies focused on system that process independently with their tasks. Likewise of CheXNet (Rajpurkar et al., 2017) achieving high accuracy scores but offer nothing elses other than detection. Models for report generations developed by Alfarghaly et al. (2021) and Yang et al. (2023) produces a report with fluent text but only independently operates. They sometimes generates report that doesnt accurately reflect the image content.

Furthermore, the challenges of this research gap will be addressed by integrating all the AI components such as ResNet-50 for detection, Grad-CAM for visual explanation, FAISS for case retrieval and Groq API for explanation. This will show how this modern AI components can be collaborating with each other into a coherent architecture that provides package like support through the components.

Moreover, most of medical AI research organizations originates from institutions with expensive set of equipment and capable of doing resource extensive research. There are few prototype that are designed for limited resource settings. This research specifically targets academic feasibility by using open-source models (ResNet-50), efficient retrieval (FAISS), and cloud-based VLM access (Groq API). At the end, the system can shows that this project can be built with accessible resource.

2.8 Summary

To sum up, this chapter has reviewed the concept of chest x-ray and each of the Artificial Intelligence (AI) modules. The synthesize of the review began by establishing the medical importance of CXR and the challenges faced by students particularly the radiologist shoetage crisis in Malaysia (Ramli & Mohd Zain, 2023).

The highlights of six target pathologies were defined including pneumonia, cardiomegaly, pleural effusion, pneumothorax, atelectasis and lung mass with their radiographic signs and clinical relevance (Ait Nasser & Akhloufi, 2023; Cho et al., 2021). The following sections critically compared the key AI technologies. ResNet-50 was selected over DenseNet-121, VGG-16 and EfficientNet for its balance of accuracy, computational efficiency and compatibility with XAI and retrieval modules (K. He et al., 2016; Showkat & Qureshi, 2022). Grad-CAM emerged as the preferred XAI method due to its clinical interpretability and validation in medical imaging (Qiu et al., 2023). FAISS was justified as the retrieval engine for its speed, scalability, and empirical use in medical imaging (Jush et al., 2024). Groq API was selected as the VLM approach for its balance of accessibility, speed, and practical deployment feasibility compared to local models like LLaVA-Med or other cloud services like GPT-4V (Wu et al., 2023; Yang et al., 2023).

Ultimately for this review revealed a worthy integration gap in current research. While individual components such as detection, explanation, retrieval, and reasoning are well-studied separately, few numbers of prototypes combine them into unified, lightweight and academically deployable systems. This research directly addresses that gap by designing and creating an end-to-end multimodal AI pipeline that demonstrates modern AI and Big Data practices on a real-world medical imaging benchmark. By treating chest X-ray analysis as a testbed for system engineering, this work contributes to research on composable, explainable, and scalable AI systems with applications extending beyond healthcare.

CHAPTER 3

RESEARCH METHODOLOGY

This chapter explains the research methodology and technical approach used to build an AI-assisted educational tool for chest X-ray interpretation. Throughout this chapter, the phases of Software Development Life Cycle (SDLC) and the application towards the project will be further explained and elaborated.

3.1 Introduction

This chapter details on the development of the project prototype that integrates multiple AI in a pipeline. This development initialize with Software Development Life Cycle (SDLC) framework which provides a structured approach to designing, implementing and evaluating complex software systems (Jackson et al., 2025). Each phases in the selected SDLC models ensures a systematic progress from initial requirement analysis through final system evaluation.

3.2 Research Framework

The framework guides the project from initial requirements identification to final system evaluation. It organized into four primary phases aligned with SDLC model such as requirements analysis, system design, implementation and testing & evaluation. Tables 3.1 provides a comprehensive summary of this research framework with the phases corresponds research objectives elaborating specific activities and outcomes produced at each phases.

3.2.1 Software Development Life Cycle (SDLC) Overview

The Software Development Life Cycle (SDLC) consist of seven phases which is planning, analysis, design, coding/implementation, testing, deployment and maintenance (Jackson et al., 2025). Each phase has specific goals that guide the progression to the next stage. Figure 3.1 illustrates the complete SDLC process.

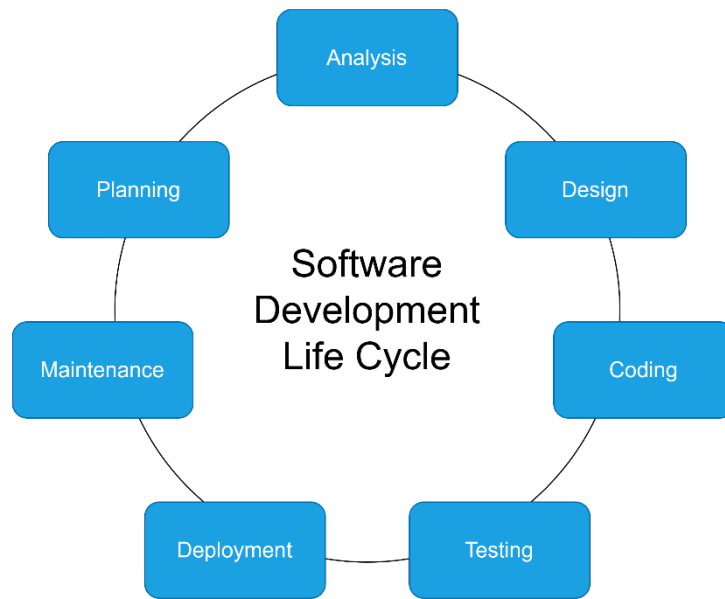


Figure 3.1: System Development Life Cycle (SDLC) diagram, adapted from (Jackson et al., 2025).

For this project, the Waterfall model was selected due to its linear and sequential structure where each phase must be completed before proceeding to the next. This predictability is suitable for research with constraints timeline and clear objectives. Figure 3.2 shows the Waterfall model adapted for this research with its phases.

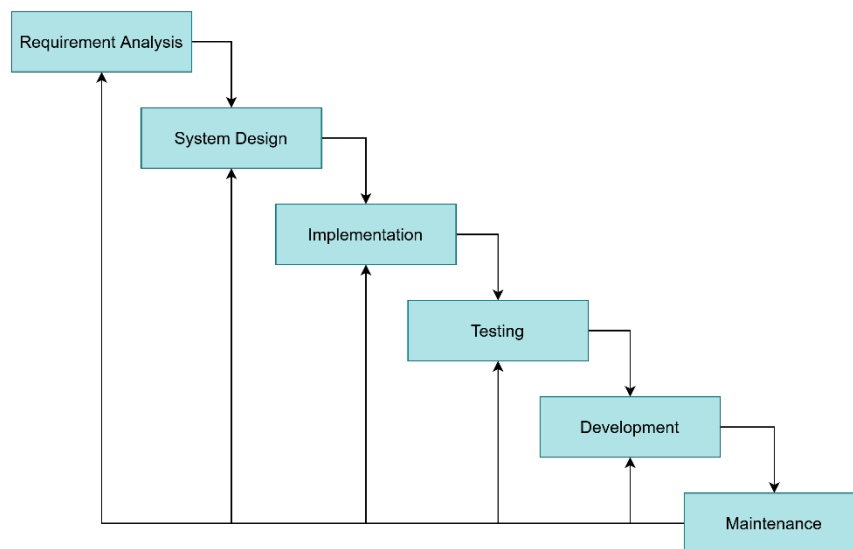


Figure 3.2: Waterfall model diagram, adapted from (Jackson et al., 2025).

In this research, only four phases will be incorporated for prototype development which is requirement analysis, system design, implementation and testing. Table 3.1 maps each research objective to specific SDLC phases, activities, and outcomes, providing a clear roadmap for the project execution.

Table 3.1
Research Framework Summary

Objective	Phases	Activity	Outcome
Objective 1: To analyze an approach and suitable method to integrate multiple Artificial Intelligence models in a modular pipeline that is able to assist medical students	Phase 1: Requirement Analysis	Discussed with the supervisor about the final year project topic	Background study, problem statement, objectives, scope, and significance
		Identified and Figured out the problem, objectives, and solutions	
		Conducted a survey using short interviews with a small number of medical students.	
		Conducted a literature review related to education and medical domains:-	Literature review report on relevant information studied for this project
		- Articles - Websites - Journal - Research Paper	
		Collected more than 750-1,000 images of chest x-ray on each type of pathologies accessed from reputable website	
Objective 2: To design and implement a modular pipeline integrating detection,	Phase 2: System Design	Designed the integrated AI model architecture system	Integrated AI model system architecture

Objective	Phases	Activity	Outcome
localization, retrieval, and explanation modules during project timeline		Designed the integrated AI model use case diagram	Integrated AI model use case diagram
		Designed the integrated AI model flowchart	Integrated AI model flowchart
		Created the integrated AI module design specification	Integrated AI module design specification
		Designed the integrated AI model entity relationship diagram	Integrated AI model entity relationship diagram
		Designed the integrated AI model user interface (UI) design	Integrated AI model user interface (UI) design
	Phase 3: Implementation	Determined what software requires for the system	Software requirements specification
		Determined what hardware requires for the system	Hardware requirements specification
		Developed the integration of the system	Fully integrated AI system with functional modules
Objective 3: To evaluate the system's performances and effectiveness in improving diagnostic accuracy via technical and usability testing	Phase 4: Testing & Evaluation	Conducted System Usability Scale (SUS) assessment with test users	SUS score and usability feedback analysis
		Tested detection module performance on held-out test set	Technical performance evaluation report with classification metrics

3.3 Phase 1: Requirement Analysis

This initial phase focused on establishing the project scope, validating functional requirements, and preparing the necessary datasets. This phase was critical to ensure that the subsequent design and implementation phases were built upon clear technical objectives and high-quality data.

3.3.1 Project Analysis

During this phases, a plan was established to concentrates the basis of the project by finding out the technical choices for interpreting chest X-ray to assists user on detecting pathologies and produces reasoning through AI tools. A literature review was conducted to validate technical choices for each system component by reviewing other publications and research paper. The literature materials was accessed from UiTM Online Database and Google Scholar that have gave shortcut for acquiring such materials. This was done to better understand and learnt various important topics and key findings like deep learning, explainable AI, vector-based case retrieval and visual language models. This topic were necessary for this research in order to evaluate what techniques and methodologies will be most effective for this research.

The outcomes of this phase is a clearly defined problem statement, objectives, scope and what significance to the research that emphasizes the system overall ability along with validated technical justifications for component selection.

3.3.2 Dataset Selection and Preparation

Data is a neccessary materials in order to extract and acquiring information for certain context. To realize this, the research will utilizes the CheXpert dataset which is a large-scale public repository of chest radiographs. This dataset can be used to train and validate the system's detection and retrieval capabilities. The dataset was selected not only for its medical applicable but specifically for its scale, labeling quality and suitability.

This dataset was accessed through the Kaggle datasets library from the website that can be accessed throught this link: <https://www.kaggle.com/datasets/ashery/chexpert>. The dataset was originally created by Stanford Hospital but it was published by ashery in Kaggle website and makes the CheXpert dataset easily accessible for research purposes.

The original CheXpert dataset are made up of backdated chest X-rays collected from patients at Stanford Hospital between October 2002 and July 2017. The dataset population consists of adult patients from both inpatient and outpatient of the hospital. This diversity

ensures the exposure to a wide range of patient demographics, clinical scenarios and image qualities that include variable positioning (AP/PA views). The dataset contains 224,316 chest radiographs of 65,240 patients with labeling and uncertainty annotations that reflect real-world diagnostic ambiguity which making it particularly valuable for training multi-label classification models.

Due to computational constraints, the final dataset will be preprocessed to consists of approximately 5,000 to 7,000 PA view images covering the six target pathologies. To ensure the reliability for evaluation, the sampling process was conducted at the patient level rather than the image level. This preventing multiple images from the same patient appearing in both training and testing sets and also preventing data leakage and bias in the evaluation.

Evaluation of the system performance will be use the collected dataset that was partitioned into two distinct subsets using a standard 80:20 ratio. The training set which is 80% of the dataset consisting of approximately 4,000 to 5,600 images is allocated for training the CNNs. This set can teaches the model to recognize specific visual features of the six target pathologies. The test set which is 20% of the dataset containing the remaining 1,000 to 1,400 images is reserved exclusively for the final system evaluation.

3.3.3 Data Preprocessing

Before the selected images could be utilized for model training, data processing were necessary to standardize the input data, removing inconsistencies and ensure the deep learning model recieves uniform input.

This preprocessing stage include image resizing, formatting and normalization. The original X-rays dataset vary significantly in resolution and aspect ratio. To make these compatible, all images were resized to a fixed dimension of 224×224 pixels. This dimension is the standard for pretrained CNNs in industry.

Medical Images differ in contrast and brightness due to variations in X-ray machine calibration. Thus, all pixel values that are originally ranging from 0 to 255 were scalled to a normalized range of 0 to 1. Furthermore, the dataset was normalized using the standard mean and standard deviation values of the ImageNet dataset (Mean: [0.485, 0.456, 0.406], Std: [0.229, 0.224, 0.225]). This step ensures that the transfer learning process remains stable.

3.4 Phase 2: System Design

Phase 2 is the first part of Research Objective 2. The design decisions are driven by the challenges identified in Phase 1, that led to creation of system architecture diagrams, use case specifications, system flowcharts, module designs, Entity Relationship Diagram (ERD) and user interface design.

3.4.1 System Architecture Overview

The system adopt a layered system architecture, its consistent with standard software engineering practices and the SDLC methodology. Figure 3.3 illustrates the complete system architecture. The layers includes Presentation Layer, Authentication Layer, Application Layer, AI Module Layer and Data Layer.

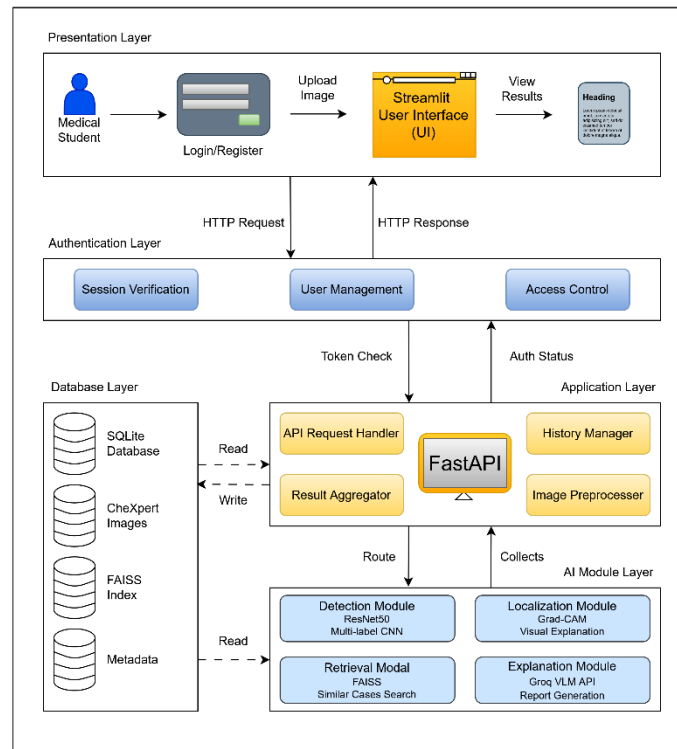


Figure 3.3: System architecture diagram

The Presentation Layer consists of a web-based user interface built with Streamlit. Here, the user login or register to uploads a chest X-ray image, which is then sent to the Application Layer via an HTTP request. In Authentication Layer, user session are verified by their login credentials. Access Control and User Management are also included to accompany the flow. In the Application Layer, a FastAPI backend receives the request and triggers preprocessing. The processed image is then routed to the AI Module Layer.

The AI Module Layer contains four core modules which is Detection Service, Localization Service, Retrieval Service and Explanation and Report Generation Service. Each

AI models function corresponds with their name and produces output respectively. The data layer will be read by AI Module Layers for its process. Also read/write process happens in Authentication Layer (for user data) and Application Layer (for history manager). The results from AI Module Layer will be collected by result aggregator before passing it back to Presentation Layer through HTTP response for displaying.

3.4.2 Use Case Diagram

To elaborate on user interactions with the system, Figure 3.4 illustrates the system's functional capabilities. The use case diagram shows how users interact with different components of the system.

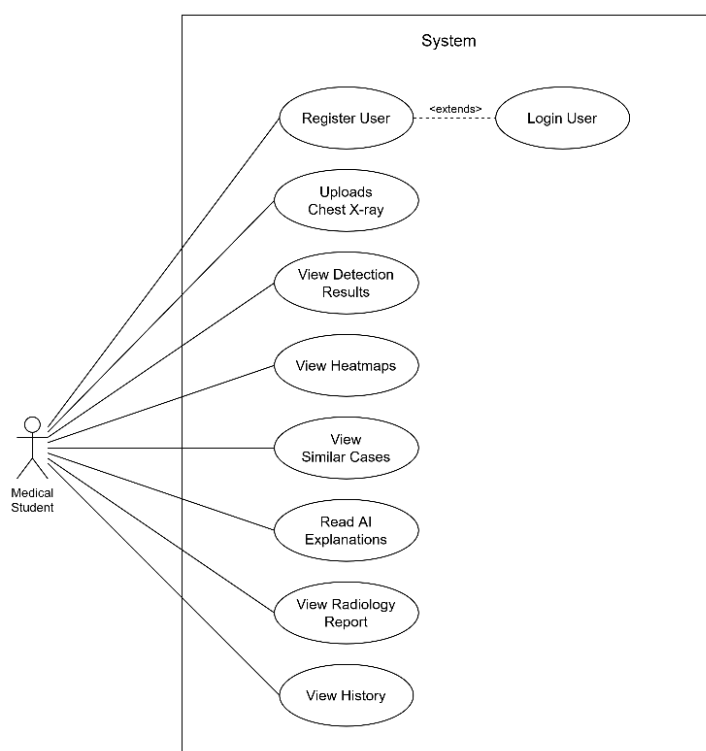


Figure 3.4: Use case diagram

As visualized in Figure 3.4, the system allows users to login or register and to input chest X-ray images and access multiple outputs from each processing stage. This includes detection results, heatmap visualizations, similar historical cases, diagnostic explanations, and structured radiology reports. These use cases are detailed in Tables 3.2 to 3.10, which describe each interaction's purpose, preconditions, postconditions, and event flows including normal, alternative, and exceptional scenarios.

Table 3.2
Login User

Use Case Name	Login User
Brief Discription	Registered user authenticates to access the system
Main Actor	Medical Student
Pre-Condition	User has a registered account and is on the login page
Post-Condition	User is authenticated and granted access to the main application
Event Flow	
Normal Flow	N1: User enters username/email and password N2: System retrieves user record from database N3: System verifies password against crypted hash N4: System creates session record in sessions table N6: System returns token to frontend and stores in session state N7: User is redirected to main application interface
Alternative Flow	A1: Incorrect username/email System displays: <i>"Invalid credentials"</i> A2: Incorrect password System displays: <i>"Invalid credentials"</i>
Exceptional Flow	E1: Database unavailable System displays: <i>"Login service temporarily unavailable"</i> E2: Token generation fails System logs error and displays: <i>"Authentication error. Please try again"</i>

Table 3.3
Register User

Use Case Name	Register User
Brief Discription	A new user creates an account to access the system
Main Actor	Medical Student
Pre-Condition	User is on the registration page and does not have an existing account
Post-Condition	New user account is created and stored in the database
Event Flow	
Normal Flow	N1: User enters username, email, and password N2: User confirms password by re-entering it N3: System validates input (email format, password strength, uniqueness) N4: System hashes password N5: System creates new record in users table N6: System displays success message and redirects to login page
Alternative Flow	A1: Invalid email format System displays: <i>"Please enter a valid email address"</i> A2: Username already exists System displays: <i>"Username is already taken"</i> A3: Email already registered System displays: <i>"An account with this email already exists"</i> A4: Weak password System displays: <i>"Password must be at least 8 characters with 1 uppercase letter and 1 number"</i> A5: Passwords don't match System displays: <i>"Passwords do not match"</i>
Exceptional Flow	E1: Database connection error System displays: <i>"Registration failed. Please try again later"</i>

Table 3.4
Upload Chest X-Ray

Use Case Name	Upload Chest X-Ray
Brief Discription	The user uploads a chest X-ray image for analysis
Main Actor	Medical Student
Pre-Condition	The user is on the system's main page
Post-Condition	Image is received and stored temporarily for processing
Event Flow	
Normal Flow	N1: User clicks the "<i>Upload</i>" button. N2: User selects a PNG or JPG file. N3: System accepts the file and displays a preview.
Alternative Flow	A1: Invalid file format System displays: "Please upload a PNG or JPG image." (Users can try again.) A2: File too large System displays: "File must be under 10 MB."
Exceptional Flow	E1: System error during upload File upload fails due to server/network issues. System displays: "<i>Upload failed. Please try again.</i>"

Table 3.5
View Detection Result

Use Case Name	View Detection Result
Brief Discription	The system displays whether any of the six target pathologies are detected
Main Actor	Medical Student
Pre-Condition	A valid chest X-ray has been uploaded and processed
Post-Condition	Detection results are displayed on screen
Event Flow	
Normal Flow	N1: System runs ResNet-50 on the image. N2: For each pathology, show “Yes” (if ≥ 0.5) or “No”. N3: Results appear in a clear list.
Alternative Flow	A1: No pathologies detected System shows: “<i>No abnormalities detected among the six target conditions.</i>”
Exceptional Flow	E1: CNN model fails to load ResNet-50 model is missing or corrupted. System logs error and shows: “<i>Detection service unavailable.</i>”

Table 3.6
View Heatmaps

Use Case Name	View Heatmaps
Brief Discription	The user views a heatmap overlay showing where the AI “sees” abnormalities
Main Actor	Medical Student
Pre-Condition	At least one pathology was detected
Post-Condition	Heatmap is displayed over the original X-ray
Event Flow	
Normal Flow	N1: System generates Grad-CAM heatmap. N2: User sees original X-ray with optional “<i>Show Heatmap</i>” toggle. N3: Heatmap highlights relevant regions in red.
Alternative Flow	A1: No pathology detected Heatmap option is disabled or hidden.
Exceptional Flow	E1: Grad-CAM computation error Heatmap generation fails (e.g., invalid tensor shape). System skips the heatmap and shows: “<i>Visual explanation unavailable for this case.</i>”

Table 3.7
View Similar Cases

Use Case Name	View Similar Cases
Brief Discription	The user views 3 past chest X-rays that are visually similar to the current case
Main Actor	Medical Student
Pre-Condition	Image has been processed by the retrieval module
Post-Condition	Three similar cases (image + report) are displayed
Event Flow	
Normal Flow	N1: System queries FAISS with image embedding. N2: Top-3 matches are retrieved. N3: Each case shows image, labels, and original radiology report.
Alternative Flow	A1: Retrieval fails (e.g., index missing) System shows: “<i>Similar cases unavailable.</i>”
Exceptional Flow	E1: FAISS index not found Vector database file is missing. System shows: “<i>Case retrieval service is offline.</i>”

Table 3.8
Read AI Explanation

Use Case Name	Read AI Explanation
Brief Discription	The user reads explanation for the AI’s diagnosis
Main Actor	Medical Student
Pre-Condition	Detection and VLM modules have completed processing
Post-Condition	Explanation text is displayed in the UI
Event Flow	
Normal Flow	N1: System sends image + labels + heatmap to VLM. N2: VLM returns diagnostic reasoning. N3: Text appears under “<i>Explanation</i>” heading.
Alternative Flow	A1: VLM fails (e.g., API timeout) System shows: “<i>Explanation temporarily unavailable.</i>”
Exceptional Flow	E1: VLM API timeout or failure Groq API is unreachable. System displays: “<i>Explanation generator is temporarily down.</i>”

Table 3.9
View Radiology Report

Use Case Name	View Radiology Report
Brief Discription	The user views a structured radiology summary of findings
Main Actor	Medical Student
Pre-Condition	VLM has generated a report.
Post-Condition	Structured reports in ‘Findings’ and ‘Impression’ format are displayed
Event Flow	
Normal Flow	N1: System parses VLM output into predefined ‘Findings’ and ‘Impression’ sections. N2: Report appears in a clean, clinical format.
Alternative Flow	A1: No structured output System displays a simplified message based on detection results.
Exceptional Flow	E1: Report parsing fails VLM returns malformed output (e.g., no “<i>Findings</i>” section). System falls back to basic detection labels and shows: “<i>Structured report not available.</i>”

Table 3.10
View History

Use Case Name	View History
Brief Discription	System records the diagnosis result for future review and learning analytics
Main Actor	Medical Student
Pre-Condition	A diagnosis result has been created and save in the records
Post-Condition	User are able to view the history of the diagnosis results created
Event Flow	
Normal Flow	N1: User selects “<i>view history</i>” button N2: System retrieves past results from records N3: System displays a list of past results
Alternative Flow	A1: Empty past results database System does not have a result history
Exceptional Flow	E2: Database read fails System displays: "<i>Failed to retrieve the queries.</i>" User can retry or continue

3.4.3 System Flowchart

The system flowchart in Figure 3.5 illustrates the complete flow of the system from resgistering, log in, interaction with the functions and log out.

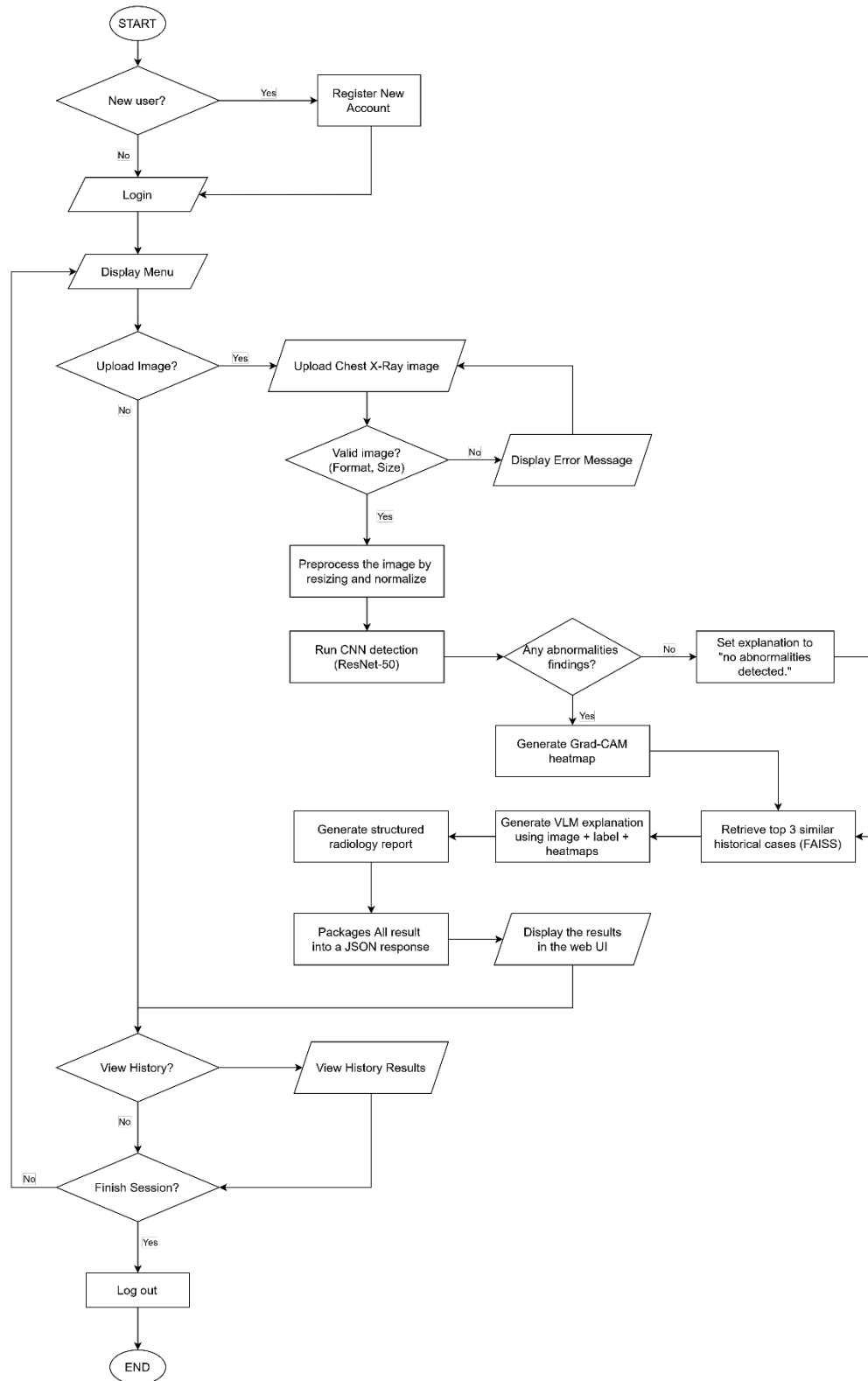


Figure 3.5: System Flowchart

The process begins when the user login into the system. If user was new, it will be directed to register page to create a new account and login afterwards. At the landing page, user will given choice to uploads a chest X-ray image via the web interface. The system first validates the image format (PNG or JPG only). If invalid, an error message appears and the user may retry. Valid images are then preprocessed by resizing to 224×224 pixels and normalizing pixel values. A ResNet-50 model performs multi-label classification to detect six possible pathologies. If no abnormalities are found, the system sets the explanation to "no abnormalities detected" and proceeds directly to case retrieval.

When abnormalities are detected, the system generates a Grad-CAM heatmap to localize findings visually. It then retrieves the top three similar historical cases using FAISS for case-based comparison. A VLM generates a detailed explanation using the image, labels, and heatmap as context. A structured radiology report is assembled using a clinical template, with content generated by the VLM for Findings and Impression sections. All results will displayed in the web UI. The user is then asked if they want to view past results of diagnosis. If yes, user will directed to list of history results. If user want to end the session, they can logout. And if not, they can upload another image to repeat the process.

3.4.4 Module Design Specification

To make the system clear and easy to test, it is divided into six separate modules. Each module has a specific job, receives certain inputs and produces specific outputs. This modular design allows each part to be developed and tested independently. If one module needs to be changed or replaced, the other modules are not affected.

This section will decribes each module in detail including Detection, Localization, Retrieval, Explanation, Authentication, User Interface and Database Management Module using Table 3.11 to 3.17.

Table 3.11
Detection Module Specification

Attribute	Description
Purpose	To identify which of the six chest diseases are present in the X-ray image.
Input	Preprocessed chest X-ray image
Output	Six probability scores each between 0.0 and 1.0 Example: `{"pneumonia": 0.92, "cardiomegaly": 0.15, "pleural_effusion": 0.78, "pneumothorax": 0.03, "atelectasis": 0.12, "lung_mass": 0.08}`
Model	ResNet-50

Operation	The model uses binary cross-entropy loss, which means it treats each disease independently. A patient can have multiple diseases at the same time. A score above 0.5 means the disease is likely present. A score below 0.5 means the disease is likely absent.
Reference	(K. He et al., 2016; Showkat & Qureshi, 2022)

Table 3.12
Localization Module Specification

Attribute	Description
Purpose	To show WHERE in the X-ray the AI model found the abnormalities.
Input	The original preprocessed image The detection results from the Detection Module The target pathology to explain (e.g., "pneumonia")
Output	A heatmap PNG file image that is overlaid on top of the original X-ray. Red/yellow areas show high importance. Blue areas show low importance.
Method	Gradient-weighted Class Activation Mapping (Grad-CAM)
Operation	Grad-CAM calculates gradients from the final prediction back to the last convolutional layer of ResNet-50. Areas with stronger gradients receive brighter colors in the heatmap.
Reference	(Qiu et al., 2023; Selvaraju et al., 2017)

Table 3.13
Retrieval Module Specification

Attribute	Description
Purpose	To support learning by finding other chest X-rays that look similar to the current one.
Input	A 2048-dimensional image embedding (feature vector) extracted from ResNet-50's global average pooling layer.
Output	Top-3 most similar historical cases. For each case the system returns: A thumbnail image preview Pathology labels The original radiology report text Similarity score
Method	FAISS (Facebook AI Similarity Search)
Operation	1. Extract embedding from the query image using ResNet-50

	2. Search the FAISS index for the 3 nearest embeddings based on L2 distance 3. Retrieve the corresponding image IDs 4. Look up full case details from the metadata
Reference	(Johnson et al., 2021; Jush et al., 2024)

Table 3.14
Explanation Module Specification

Attribute	Description
Purpose	To generate natural language explanations that describe what the AI sees in the X-ray and why it made certain diagnosis.
Input	The original chest X-ray image Detection results (e.g., " <i>Pneumonia: Yes</i> ") Grad-CAM heatmap
Output	1. Natural Language Explanation: Example: " <i>The right lower lobe shows focal consolidation with air bronchograms, consistent with bacterial pneumonia.</i> " 2. Structured Radiology Report: Findings that describes what is visible and impression that states the likely diagnosis with supporting evidence
Model	Vision Language Model (VLM) accessed through Groq API.
Operation	1. The image is encoded as base64 text for API transmission 2. First API call: Generate the Findings section 3. Second API call: Generate the Impression section using the Findings as context 4. The system combines both sections into a standard radiology report format
Reference	(Groq, 2024; Yang et al., 2023)

Table 3.15
Authentication Module Specification

Attribute	Description
Purpose	To manage user accounts and control access to the system. Only registered and logged-in users can use the AI analysis features and view the analysis history.
Components	1. Registration (Create new user accounts) 2. Login (Verify credentials and create sessions) 3. Session Verification (Check if user is logged in for every request)
Input	Registration is username, email and password Login is username and password

	Verification is session token from request cookie
Output	Registration is user_id (successful) or error message (failed) Login is session token (successful) or error message (failed) Verification is user_id (valid session) or error (invalid/expired)
Operation	Registration validate and check if the credentials is in format and if already in database. Bcrypt password and store it in database. Login Process Verify if username is in database and password matches the hashes. Generate and store session token with timestamp and return it to user. Verification extract token and look up it session. Check last activity timestamp and give user authentication to proceed.
Security Features	Passwords hashed with bcrypt Session tokens generated using cryptographically secure random generator Sessions expire after 24 hours of inactivity

Table 3.16
User Interface Module Specification

Attribute	Description
Purpose	To provide a simple and easy-to-use web interface.
Technology	Streamlit
Main Page	Login/Register Page Main Analysis Page History View Page
Design	Simplicity: Only essential features, no complicated menus Clarity: Large text, clear headings, organized sections Feedback: Immediate response to user actions Accessibility: High contrast colors, readable fonts Focus: Emphasis on learning, not fancy graphics
Reference	(Sait & Tombs, 2021)

Table 3.17
Database Management Module Specification

Attribute	Description
Purpose	To store and manage all user data, including accounts, login sessions and interpretation history. Also provides access to chest X-ray metadata.
Storage Type	SQLite Database (Read-Write) Users table: User accounts and credentials Sessions table: Active login sessions

	Interpretation_history table: Diagnostic attempts chexpert_metadata: Chest X-ray labels and reports Image files: 5,000-7,000 PNG images
Database Location	Single file: `data/chestxray_ai.db`
Operation	For User Data: CREATE: Add new user accounts READ: Check login credentials, retrieve history UPDATE: Refresh session activity time DELETE: Remove expired sessions, delete accounts For X-ray Data: READ ONLY: Load images and metadata
Methods	SQLAlchemy ORM: All database operations use Python objects Pandas: Load CSV metadata into memory once at startup PIL: Load image files when needed

3.4.5 User Interface (UI) Design

User Interface (UI) Design is crucial to illustrate how the user will be able to see and interact with the interface. Below is the design of the UI of the system that uses a web platform using Figure 3.6 to 3.10.

The image shows a web browser window with several tabs. The active tab is titled 'streamit' and shows the URL 'www.streamit.com/register'. The page content is a registration form with the title 'Register' in bold. The form includes four text input fields labeled 'Username', 'Email Address', 'Password', and 'Re-Enter Password'. At the bottom of the form is a green button labeled 'Submit'.

Figure 3.6: System's Register Page

Figure 3.6 shows the Register Page for user to create a new account. This registration is necessary in order to allow user to login with the account and access the feature of the system. From the figure, user required to input credentials and submit it to register a new account.

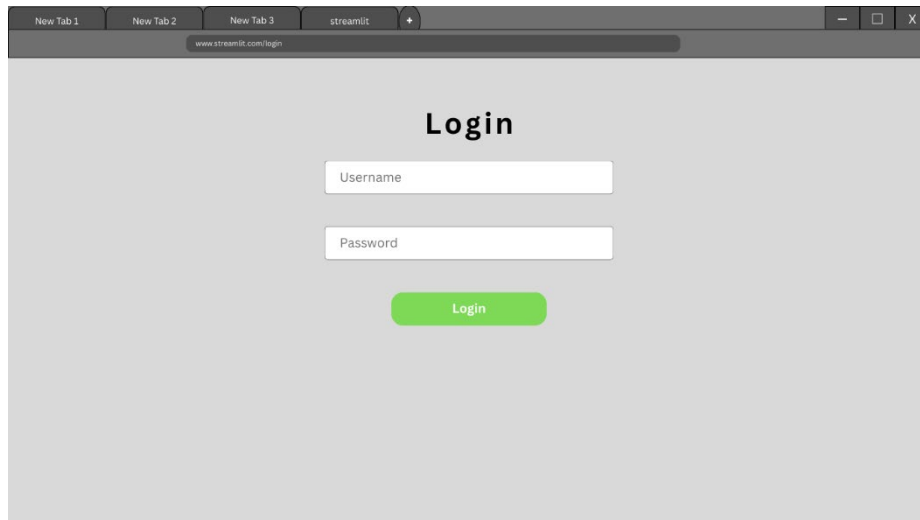


Figure 3.7: System's Login Page

Figure 3.7 shows the Login Page for user to access the system. User required to input their account credentials and click the button “Login” to proceed towards the system Home Page.

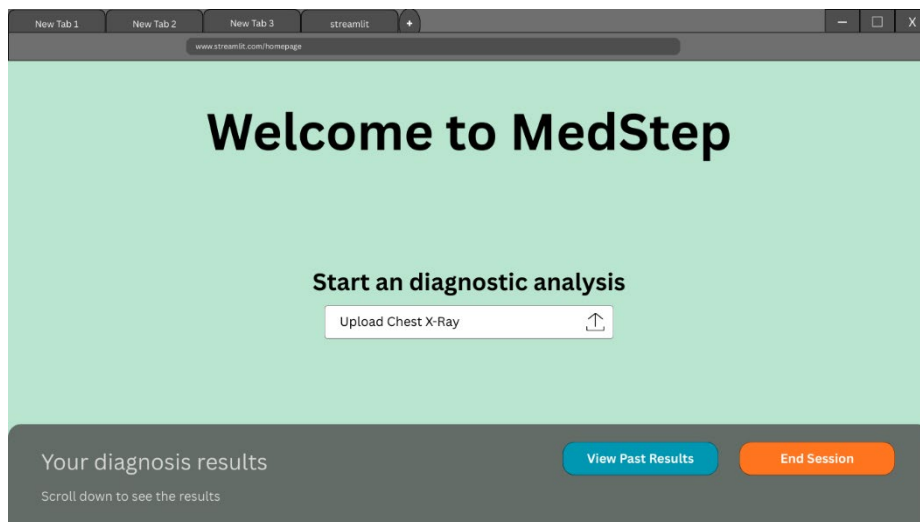


Figure 3.8: System's Main Page

Figure 3.8 shows the Main Page as the landing page after user recent login. This page gives user options whether to upload a chest X-ray image to start the diagnostic analysis, View past results or end the current session. Here if user initiates the diagnostic analysis process, user will be able to view their results at the below of Main Page.

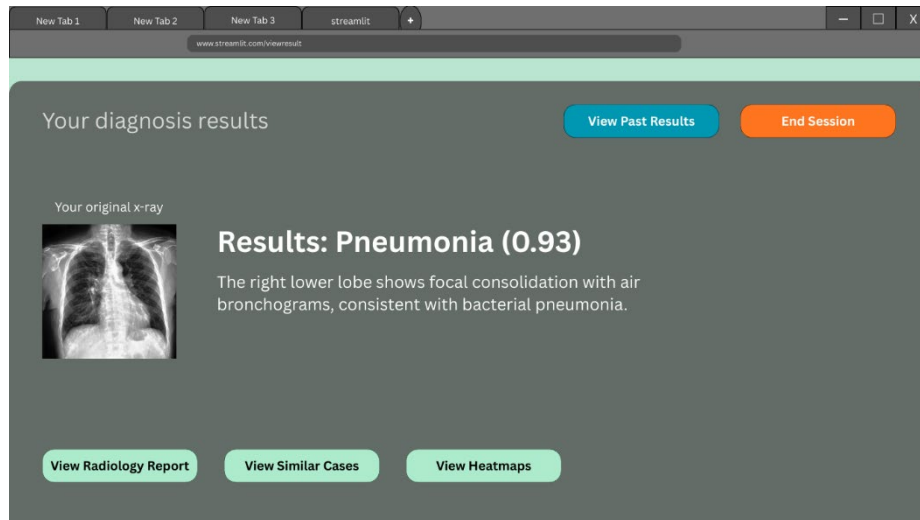


Figure 3.9: System's View Result Page

Figure 3.9 show the View Result Page for user to able sees their diagnosis result. The system will display their original photos, the pathologies detected and its accuracy, the findings and impression summary. User also gave ability to sees radiology report, similar cases and heatmaps by pressing the related button below the main diagnosis results.

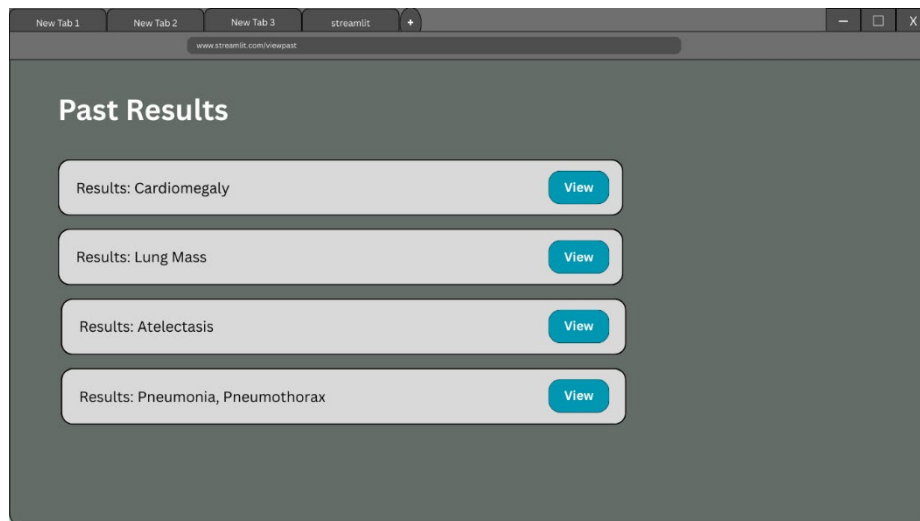


Figure 3.10: System's View Past Results Page

Figure 3.10 show the View Past Results Page which include a list of user attempted diagnosis results created from this system. User were displayed with their results including the label of the pathologies. A button “view” also featured for detailed results display.

3.4.6 Database Design

This research will implement database management system using SQLite and SQLAlchemy. It's due to it lightweight and easy to use as it was part of Python programming. This designs allow to keep balances the simplicity required for prototypes with the functionality needed for personalized features while maintaining the integrity of the existing AI processing pipeline.

3.4.6.1 Entity Relationship Diagram (ERD)

The database schema consists of four tables that includes users, session, history, and CXR_case table through logical foreign key relationships. Figure 3.11 illustrates the complete data model.

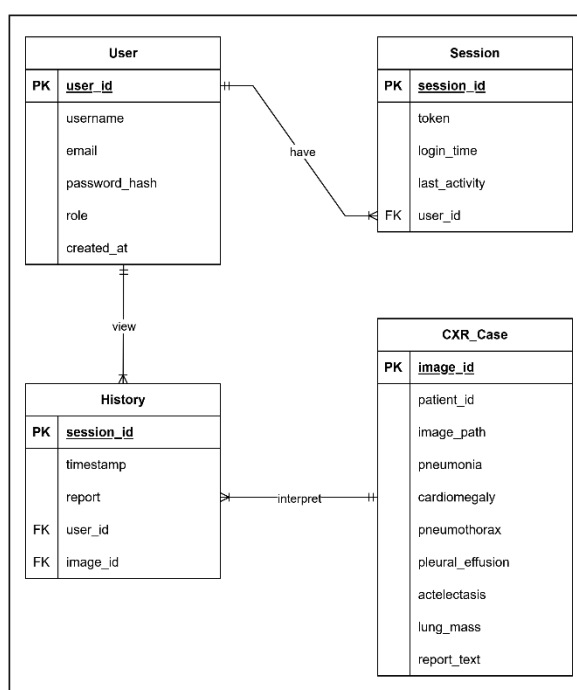


Figure 3.11: System Entity Relationship Diagram

3.4.6.2 Data Dictionary

The system's metadata is stored in a structured SQLite database. Each row represents one chest X-ray case. Table 3.18 to 3.21 defines all fields used in this file.

Table 3.18
Data Dictionary of table Users

Table name: Users				
Field Name	Data Type	Constraints	Description	Example
user_id	INTEGER	PRIMARY KEY, AUTO INCREMENT, NOT NULL	Unique identifier for each user account. Automatically generated on registration	1
username	VARCHAR(50)	UNIQUE, NOT NULL	User's login username. Must be 3-50 characters, alphanumeric only. Case-insensitive for login but preserved as entered	"student123"
email	VARCHAR(100)	UNIQUE, NOT NULL	User's email address. Must match basic email format (contains @ and domain)	"2025123456@student.uitm.edu.my"
password_hash	VARCHAR(255)	NOT NULL	Bcrypt hash of user's password. Never stores plaintext. Hash includes automatically generated salt.	"\$2b\$12\$abcd..."
role	VARCHAR(20)	NOT NULL, DEFAULT 'student'	User's access level.	"student"
created_at	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Account creation date and time in UTC. Automatically set on registration.	"2026-03-15 10:30:00"

Table 3.19
Data Dictionary of table Session

Table name: Session				
Field Name	Data Type	Constraints	Description	Example
session_id	INTEGER	PRIMARY KEY, AUTO INCREMENT, NOT NULL	Unique identifier for each login session	1

token	VARCHAR(32)	UNIQUE, NOT NULL	Cryptographically secure random session token. Generated using Python's `secrets.token_urlsafe(32)`. Sent with every request in Authorization header.	"a3k9mP2nQ8rT5vW..."
login_time	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When user successfully authenticated (UTC). Used for session duration tracking.	"2026-03-15 10:30:00"
last_activity	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Most recent API request timestamp (UTC). Updated on every request. Used to determine session expiration (24 hours inactivity)	"2026-03-15 14:45:30"
user_id	INTEGER	FOREIGN KEY (users.user_id), NOT NULL	Links session to user account. Enables querying all sessions for a user	1

Table 3.20
Data Dictionary of table History

Table name: History				
Field Name	Data Type	Constraints	Description	Example
history_id	INTEGER	PRIMARY KEY, AUTO INCREMENT, NOT NULL	Unique identifier for each interpretation attempt	1
timestamp	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When diagnosis results attempted (UTC). Used for chronological ordering and progress tracking over time	"2024-03-15 10:45:30"
report	TEXT	NULL	Diagnosis results explanation.	"Looked at right lower lobe, saw opacity..."
image_id	VARCHAR(100)	FOREIGN KEY (CXR_Case.image_id), NOT NULL	Links to specific chest X-ray case from CheXpert dataset. Enables tracking which cases were reviewed	"patient123_study1_view1"

user_id	INTEGER	FOREIGN KEY (users.user_id), NOT NULL	Links interpretation to user account. Enables filtering history by user	1
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Table 3.21
Data Dictionary of table CXR_case

Table name: CXR_case				
Field Name	Data Type	Constraints	Description	Example
image_id	VARCHAR(100)	PRIMARY KEY, UNIQUE, NOT NULL	Unique identifier for each chest X-ray image. Used as filename prefix and foreign key in interpretation_history	"patient123_study1_view1"
patient_id	VARCHAR(50)	NOT NULL	Anonymous patient identifier from CheXpert dataset. Used during dataset preparation to prevent data leakage (same patient's images don't appear in both train and test sets)	"patient00123"
image_path	VARCHAR(255)	NOT NULL	Relative file path from project root to preprocessed PNG image file	"data/images/patient123_study1_view1.png"
pneumonia	INTEGER	NOT NULL	Pathology label from CheXpert. Values: 1 = positive, 0 = negative, -1 = uncertain. Uncertain labels are excluded during model training	1, 0, -1
cardiomegaly	INTEGER	NOT NULL	Same labeling convention as pneumonia	1, 0, -1
pleural_effusion	INTEGER	NOT NULL	Same labeling convention as pneumonia	1, 0, -1
pneumothorax	INTEGER	NOT NULL	Same labeling convention as pneumonia	1, 0, -1
atelectasis	INTEGER	NOT NULL	Same labeling convention as pneumonia	1, 0, -1
lung_mass	INTEGER	NOT NULL	Same labeling convention as pneumonia	1, 0, -1

report_text	TEXT	NULL	Original radiology report from CheXpert dataset. Anonymized (patient names removed). Used for displaying in similar cases retrieval	"Findings: Right lower lobe consolidation. Impression: Pneumonia"
embedding_index	INTEGER	NOT NULL	Position of this image's 2048-dimensional feature vector in FAISS index file. Used for efficient similarity search lookup	0, 1, 2, ...

3.5 Phase 3: Implementation

Phase 3 addresses the implementation component of Research Objective 2 by translating the system design from Phase 2 into a functional prototype. This phase will focus on selecting which appropriate software and hardware resources for implementing each AI module according to the specifications and integrates the components into a unified pipeline.

3.5.1 Software Requirements

The system requires certain software for the implementation. This includes a combination of programming languages, frameworks and libraries that are selected by their performance, compatibility and massive community support. All of the software selected is open source or freely accessible. Table 3.22 lists the essential software requirements.

Table 3.22
Software Requirements

Category	Component	Version	Purpose
Programming Language	Python	3.8+	Primary language for all system components.
Deep Learning	PyTorch	2.0+	Neural network training and inference for ResNet-50 model.
	TorchVision	0.15+	Pretrained ResNet-50 models and preprocessing utilities.
Web Interface	Streamlit	1.28+	User interface development for web-based interaction.
	FastAPI	0.104+	Backend API server for coordinating AI modules.
Explainable AI	Pytorch-grad-cam	1.4+	Grad-CAM implementation for visual explanations.
Vector Retrieval	FAISS	1.7+	Efficient similarity search of image embeddings.
Visual Language Model	Groq SDK	0.4+	API client for accessing VLM via Groq cloud service. Provides access to vision-capable language models for report generation.
Database	SQLite	3.35+	Lightweight relational database for user data storage
	SQLAlchemy	2.0+	Object-Relational Mapping for database operations

Core Libraries	NumPy	1.24+	Numerical computing and array operations.
	Pandas	2.0+	Metadata management and CSV handling.
	Pillow (PIL)	10.0+	Image loading and format conversion.

Operating System: Windows 11

Groq API key (free tier available at <https://groq.com>)

3.5.2 Hardware Requirements

The system's computational demands are determined primarily by the deep learning model inference and vector similarity search operations. While the system is designed to run on standard academic hardware, GPU acceleration significantly improves performance, particularly for batch processing during preprocessing and real-time inference during user interactions. Table 3.23 specifies the minimum and recommended hardware configurations.

Table 3.23
Hardware Requirement

Hardware Component	Minimum Specification	Purpose
Processor (CPU)	Intel Core i5 (8th gen) or AMD Ryzen 5 (3000 series)	Handles data preprocessing, API requests, and CPU-based computations.
Graphics Card (GPU)	NVIDIA GTX 1660 (6GB VRAM)	Accelerates ResNet-50 inference, Grad-CAM computation, and FAISS similarity search.
Memory (RAM)	16GB DDR4	Prevents memory bottlenecks during concurrent operations.
Storage	100GB available space (SSD recommended)	Stores CheXpert dataset subset, pretrained models, FAISS index, code and dependencies.
Network	Stable broadband connection (10 Mbps+)	Needed for VLM API calls if using cloud-based services.
Display	1080p Full HD	Required for displaying the results.

3.5.3 Module and System Integration Development

This section will describes how each module was setup and develop together to form the complete system. The development will initiates by build each module separately, test it distinctly and then integrate all modules into a system.

3.5.3.1 Detection Module Implementation

The Detection Module uses a pretrained ResNet-50 model. This model was then fine-tuned specifically for chest X-ray analysis. ResNet-50 will be loaded using PyTorch's torchvision library. Furthermore, the output layer will be modified to output 6 probabilities (pathologies). The fine-tune process will be based on the chest X-ray dataset (CheXpert) for 10 to 25 epoch. Moreover, an optimizer will be applied for training settings and finally save the trained model for reuse.

3.5.3.2 Localization Module Implementation

The Localization Module uses Grad-CAM to generate heatmaps. This model will be used for visualization of the Detection Model results. To employ this model, pytorch-grad-cam library will be used for installation. Furthermore, its getting configuration on the target layer to extract features from ResNet-50. This model will generate separate heatmaps for each detected disease. Moreover, the heatmap will be overlaid on top of the original X-ray images with filter (40% transparency). Finally. It will be save as PNG file for temporary directory.

3.5.3.3 Retrieval Module Implementation

The Retrieval Module uses FAISS to quickly find similar chest X-ray cases. The process is by running all training images through ResNet-50 and saved the 2048-dimensional feature vectors in order to extract image embeddings. Furthermore, FAISS IndexFlatL2 index will be created and add all embeddings on to it. Then, the index will be saved and finally a function that takes a new image, extract its embeddings and searches the index will be created.

3.5.3.4 Explanation Module Implementation

The Explanation Module uses Groq's cloud API to generate natural language reports for explanation. The acquisition of Groq API key will be needed to setup the API access. Then, it is required to install Groq Python SDK using `pip install groq`. Furthermore, a default structured prompts will be created which is Findings and Impression. The chest X-ray image will be converted to base64 format for API transmission. Finally, an API call will be made to generate complete report.

3.5.3.5 Authentication, Web Interface, Database Setup

The Authentication Module manages user accounts and login sessions. The Database Models, SQLAlchemy, will define for 'users' and 'sessions' tables. Password hashing were implemented toward registration function using Passlib's brypt. The login function will be created for credential verification and token generation. Then, the token will be check on evey request.

The Web Interface was built using Streamlit for the frontend and FastAPI for the bacakend. From Streamlit, all the pages from UI design will be built. FastAPI is used for API routes of all operations. A session management store user token in the session state.

The database was set up using SQLAlchemy to automatically create tables. This process will be connected to SQLite database file for each table that were created (users, session, interpretation_history, CXR_cases). A test queries are used to verify all CRUD operations if it work correctly.

3.5.3.6 System Integration Process

After all the individual modules were built, they were integrated one by one. First, ResNet-50 and Grad-CAM. Second, ResNet-50 with FAISS. Third, detection result and heatmaps will combined to be received by Groq API. Then, database and authentication will follow through. At the end, the integrated system will be validated to make sure it run correctly.

3.6 Phase 4: Testing & Evaluation

Phase 4 addresses Research Objective 3 by evaluating the integrated system's performance through technical metrics and usability assessment. This phase will conduct a multiple testing approach including unit testing for each individual components and usability evaluation.

3.6.1 Testing Strategy Overview

The research planned a testing strategy by evaluating small components first and progressively to complete system evaluation. This ensures that every errors identified and resolved at component level. Table 3.24 summarizes the testing types, focus, methods and success criteria.

Table 3.24
Summary of testing strategy

Testing Type	Focus	Method	Success Criteria
Unit Testing	Individual module functionality	Automated tests with sample inputs	All modules produce expected outputs within specifications
Usability Testing	User interface and experience	System Usability Scale (SUS) questionnaire	SUS score is above average usability, ≥ 65

3.6.2 Unit Testing

Unit testing validates each module including Detection, Localization, Retrieval and Explanation Module using appropriate metrics that reflect its intended functionality.

3.6.2.1 Detection Module Testing

The Detection Module can be done by evaluating the performance and accuracy of the CNNs model. Various evaluation metrics can be conducted such as Accuracy, Precision, Recall and F1-Score. These different evaluation metrics are used in the module testing to determine its strengths and weaknesses.

The simplest approaches for testing the model is through Confusion Matrix. Its consists of True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN). Each of them represents on how they correctly or incorrectly predicted the whether the class positive or negative. These label are used for the evaluation metrics by applying it on their respective Equations (3.1) to (3.4).

$$Accuracy = \frac{TN + TP}{TN + FP + TP + FN} \quad (3.1)$$

$$Precision = \frac{TP}{FP + TP} \quad (3.2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3.3)$$

$$F1_Score = \frac{Precision \times Recall}{Precision + Recall} \quad (3.4)$$

The success criteria for this testing is by achieving minimum of 0.70 for F1-score for each of the pathology.

3.6.2.2 Localization Module Testing

The module, Grad-CAM, heatmaps are evaluated by its quality by verifying the highlighted regions correspond to relevant area of each pathology. This assessment involves visual inspection of a random sample of 5 cases per pathology.

The evaluation criteria is defined for each pathology. For pneumonia, the heatmap should highlight affected lung zones. For cardiomegaly, the heatmap should focus on cardiac silhouette borders. For pleural effusion, the heatmap should indicate costophrenic angles and lower lung fields. For pneumothorax, the heatmap should mark lung edges and peripheral areas. For atelectasis, the heatmap should show collapsed lung regions. For lung mass, the heatmap should localize focal opacity.

Success criteria for this testing is by having more than 80% of heatmaps correctly highlighting the relevant regions of their respective pathologies.

3.6.2.3 Retrieval Module Testing

The FAISS will be evaluated based on the speed in retrieving the similar cases. In this case, the top-3 retrieved cases are examined whether its time required to query FAISS index and return top-3 cases is fast enough. The success criteria for this testing is the retrieval speed is less than 1 second per query.

3.6.2.4 Explanation Module Testing

The module will be tested for factual accuracy and appropriate use of medical terminology. A sample of 30 generated reports is manually reviewed.

The evaluation criteria for the test is first for factual accuracy, the findings described in the report must match the detection results. Second for terminology correctness, medical terms in the report are used correctly and appropriately.

Success criteria for this testing is acquiring more than 85% of reports are factually accurate and more than 90% follow proper medical terms.

3.6.3 System Usability Scale (SUS) Assessment

The usability testing done to evaluates the UI and overall user experience with potential end user. This assessment uses System Usability Scale (SUS) that consists of 10 questionnaire item that provides a quantitative measure.

The SUS questionnaire consist of statements that to be rated on a 5-point Likert scale (1 for Strongly Disagree to 5 for Strongly Agree). To reduce the bias of the response, the questions will be alternated between postive and negative phrases. The SUS score will calculated using a formula that converts the Likert scale score to a 0-100 score.

The formula is defined by Equation (3.5) to (3.7). Each questions were calculate by subtract 1 from user's reponse if it odd number and subtract user's response from 5. Then sum all of question scores together with range from 0 to 40. Finally, multiply the sum by 2.5 to get SUS score out of 100.

$$Score (Odd Question) = (User Response) - 1 \quad (3.5)$$

$$Score (Even Question) = 5 - (User Response) \quad (3.6)$$

$$SUS Score = \left(\sum_{i=1}^{10} ItemScore \right) \times 2.5 \quad (3.7)$$

The success criteria for this testing is to achieve SUS score with more than equal to 65, which represents above average usability (acceptable) while scores more than equal to 80 represent excellent usability.

3.7 Summary

This chapter has detailed the research methodology for devoloping and evaluating the system for chest X-ray analysis. The approach was followed with the SDLC using the Waterfall model that progress systematically through four phase which is Requirement Analysis, Design Analysis, Implementation and Testing & Evaluation. The initial phase established the foundation of the research by conducting literature review to justify the selection of components and preparing a dataset for the model training.

The following design phase produces system design which include system architecture diagram, use case diagram, system flowchart, detailed module specification, ERD & data dictionary and UI for the website. Furthermore, implementation phases demonstrate by specifiying the software requirements and hardware requirements. Then, the module implemented according to the specification for the system. This coordination integrates the components into a unified pipeline to develop a system.

Toward the ends, testing phases conducted to evaluates the system performance and usability with metrics and methods. Each of the testing were focuses to achieve the success criteria in order to presents a acceptable and working system.

REFERENCES

- Ait Nasser, A., & Akhloufi, M. A. (2023). A Review of Recent Advances in Deep Learning Models for Chest Disease Detection Using Radiography. *Diagnostics*, 13(1), 159. <https://doi.org/10.3390/diagnostics13010159>
- Al Elq, A., Alfayez, A. A., AlQahtani, M. I., Alshahrani, R. S., Alotaibi, G. A., Aldakheel, A. A., Alhammad, A. A., Bakr Mohamed, R. H., Jarrar, M., Abusalah, M. A. H. A., & Al-Bsheish, M. (2023). The Effects of Various Teaching Methods on Chest X-ray Interpretation Skills Among Medical Students and Interns: A Systematic Review. *Cureus*. <https://doi.org/10.7759/cureus.44399>
- Alfarghaly, O., Khaled, R., Elkorany, A., Helal, M., & Fahmy, A. (2021). Automated radiology report generation using conditioned transformers. *Informatics in Medicine Unlocked*, 24. <https://doi.org/10.1016/j.imu.2021.100557>
- Bernhardsson, E. (2013, February 20). *GitHub - spotify/annoy: Approximate Nearest Neighbors in C++/Python optimized for memory usage and loading/saving to disk*. GitHub. <https://github.com/spotify/annoy>
- Bharati, S., Podder, P., & Mondal, M. R. H. (2020). Hybrid deep learning for detecting lung diseases from X-ray images. *Informatics in Medicine Unlocked*, 20. <https://doi.org/10.1016/j.imu.2020.100391>
- Cho, Y., Kim, J. S., Lim, T. H., Lee, I., & Choi, J. (2021). Detection of the location of pneumothorax in chest X-rays using small artificial neural networks and a simple training process. *Scientific Reports*, 11(1). <https://doi.org/10.1038/s41598-021-92523-2>
- Douze, M., Guzhva, A., Deng, C., Johnson, J., Szilvasy, G., Mazaré, P. E., Lomeli, M., Hosseini, L., & Jégou, H. (2025). THE FAISS LIBRARY. *IEEE Transactions on Big Data*. <https://doi.org/10.1109/TBDATA.2025.3618474>
- Elastic. (2023). *kNN search in Elasticsearch | Elastic Docs*. Elastic. <https://www.elastic.co/docs/solutions/search/vector/knn>
- Groq. (2024). *Overview - GroqDocs*. Groq. <https://console.groq.com/docs/overview>
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, 2016-December*, 770–778. <https://doi.org/10.1109/CVPR.2016.90>
- He, L., Luan, L., & Hu, D. (2025). Deep learning-based image classification for AI-assisted integration of pathology and radiology in medical imaging. *Frontiers in Medicine*, 12. <https://doi.org/10.3389/fmed.2025.1574514>
- Hegazi, T., Kurdi, K., Alfayez, A., Alhammad, A., Aldakheel, A., Alshahrani, R., Alotaibi, G., AlQahtani, M., Jarrar, M., Ola, H. A. Al, & Elq, A. Al. (2025). Clinical-year Students' Competency in Chest X-ray Interpretation: A Theoretical-based Intervention. *Saudi Journal of Medicine and Medical Sciences*, 13(2), 133–141. https://doi.org/10.4103/sjmms.sjmms_623_24
- Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. *Proceedings - 30th IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017, 2017-January*, 2261–2269. <https://doi.org/10.1109/CVPR.2017.243>

- Irvin, J., Rajpurkar, P., Ko, M., Yu, Y., Ciurea-Ilcus, S., Chute, C., Marklund, H., Haghighi, B., Ball, R., Shpanskaya, K., Seekins, J., Mong, D. A., Halabi, S. S., Sandberg, J. K., Jones, R., Larson, D. B., Langlotz, C. P., Patel, B. N., Lungren, M. P., & Ng, A. Y. (2019). CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison. *AAAI 2019*, 19. <https://doi.org/https://doi.org/10.48550/arXiv.1901.07031>
- Jackson, G., Kosinski, M.), & Holdsworth, J. (2025, November 17). *What is the Software Development Lifecycle (SDLC)?* | IBM. International Business Machines Corporation (IBM). <https://www.ibm.com/think/topics/sdlc>
- Jimah, B. B., Baffour Appiah, A., Sarkodie, B. D., & Anim, D. (2020). Competency in Chest Radiography Interpretation by Junior Doctors and Final Year Medical Students at a Teaching Hospital. *Radiology Research and Practice*, 2020, 1–7. <https://doi.org/10.1155/2020/8861206>
- Johnson, J., Douze, M., & Jegou, H. (2021). Billion-Scale Similarity Search with GPUs. *IEEE Transactions on Big Data*, 7(3), 535–547. <https://doi.org/10.1109/TBDATA.2019.2921572>
- Jush, F. K., Truong, T., Vogler, S., & Lenga, M. (2024). Medical Image Retrieval Using Pretrained Embeddings. *Proceedings - International Symposium on Biomedical Imaging*. <https://doi.org/10.1109/ISBI56570.2024.10635170>
- Li, C., Wong, C., Zhang, S., Usuyama, N., Liu, H., Yang, J., Naumann, T., Poon, H., & Gao, J. (2023). LLaVA-Med: Training a Large Language-and-Vision Assistant for Biomedicine in One Day. *ArXiv (Cornell University)*. <https://doi.org/https://doi.org/10.48550/arXiv.2306.00890>
- Li, M., Jiang, Y., Zhang, Y., & Zhu, H. (2023). Medical image analysis using deep learning algorithms. *Frontiers in Public Health*, 11. <https://doi.org/10.3389/fpubh.2023.1273253>
- Mould, R. F. (2018). A century of X-rays and radioactivity in medicine: With emphasis on photographic records of the early years. *A Century of X-Rays and Radioactivity in Medicine: With Emphasis on Photographic Records of the Early Years*, 1–217. <https://doi.org/10.1201/9781315136271>
- Plackett, R., Kassianos, A. P., Mylan, S., Kambouri, M., Raine, R., & Sheringham, J. (2022). The effectiveness of using virtual patient educational tools to improve medical students' clinical reasoning skills: a systematic review. *BMC Medical Education*, 22(1). <https://doi.org/10.1186/s12909-022-03410-x>
- Plesner, L. L., Müller, F. C., Brejnbøl, M. W., Laustrop, L. C., Rasmussen, F., Nielsen, O. W., Boesen, M., & Brun Andersen, M. (2023). Commercially Available Chest Radiograph AI Tools for Detecting Airspace Disease, Pneumothorax, and Pleural Effusion. *Radiology*, 308(3). <https://doi.org/10.1148/radiol.231236>
- Qiu, Z., Rivaz, H., & Xiao, Y. (2023). Is visual explanation with Grad-CAM more reliable for deeper neural networks? a case study with automatic pneumothorax diagnosis. *ArXiv (Cornell University)*. <http://arxiv.org/abs/2308.15172>
- Rajpurkar, P., Irvin, J., Zhu, K., Yang, B., Mehta, H., Duan, T., Ding, D., Bagul, A., Langlotz, C., Shpanskaya, K., Lungren, M. P., & Ng, A. Y. (2017). CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning. *ArXiv (Cornell University)*. <https://doi.org/https://doi.org/10.48550/arXiv.1711.05225>

- Ramli, N. M., & Mohd Zain, N. R. (2023). The Growing Problem of Radiologist Shortage: Malaysia's Perspective. In *Korean Journal of Radiology* (Vol. 24, Issue 10, pp. 936–937). Korean Radiological Society. <https://doi.org/10.3348/kjr.2023.0742>
- Sait, S., & Tombs, M. (2021). Teaching medical students how to interpret chest x-rays: The design and development of an e-learning resource. *Advances in Medical Education and Practice*, 12, 123–132. <https://doi.org/10.2147/AMEP.S280941>
- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. *Proceedings of the IEEE International Conference on Computer Vision, 2017-October*, 618–626. <https://doi.org/10.1109/ICCV.2017.74>
- Shin, H., Lee, J., Eo, T., Jun, Y., Kim, S., & Hwang, D. (2020). The Latest Trends in Attention Mechanisms and Their Application in Medical Imaging. *Journal of the Korean Society of Radiology*, 81(6), 1305. <https://doi.org/10.3348/jksr.2020.0150>
- Showkat, S., & Qureshi, S. (2022). Efficacy of Transfer Learning-based ResNet models in Chest X-ray image classification for detecting COVID-19 Pneumonia. *Chemometrics and Intelligent Laboratory Systems*, 224. <https://doi.org/10.1016/j.chemolab.2022.104534>
- Simonyan, K., & Zisserman, A. (2015). Very Deep Convolutional Networks for Large-Scale Image Recognition. *ArXiv (Cornell University)*. <https://doi.org/10.48550/arXiv.1409.1556>
- Sugi, M. D., Kennedy, T. A., Shah, V., & Hartung, M. P. (2021). Bridging the gap: interactive, case-based learning in radiology education. *Abdominal Radiology*, 46(12), 5503–5508. <https://doi.org/10.1007/s00261-021-03147-z>
- Tan, M., & Le, Q. V. (2020). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. *ICML 2019*. <https://doi.org/10.48550/arXiv.1905.11946>
- Thomas, A. M. K., Banerjee, A. K., & Busch, U. (2005). Classic papers in modern diagnostic radiology. *Classic Papers in Modern Diagnostic Radiology*, 1–672. <https://doi.org/10.1007/B138427/COVER>
- Wu, C., Lei, J., Zheng, Q., Zhao, W., Lin, W., Zhang, X., Zhou, X., Zhao, Z., Zhang, Y., Wang, Y., & Xie, W. (2023). Can GPT-4V(ision) Serve Medical Applications? Case Studies on GPT-4V for Multimodal Medical Diagnosis. *ArXiv (Cornell University)*. <http://arxiv.org/abs/2310.09909>
- Yang, S., Wu, X., Ge, S., Zheng, Z., Zhou, S. K., & Xiao, L. (2023). Radiology report generation with a learned knowledge base and multi-modal alignment. *Medical Image Analysis*, 86. <https://doi.org/10.1016/j.media.2023.102798>